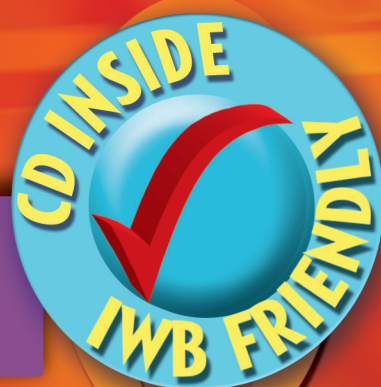


ACHIEVE!

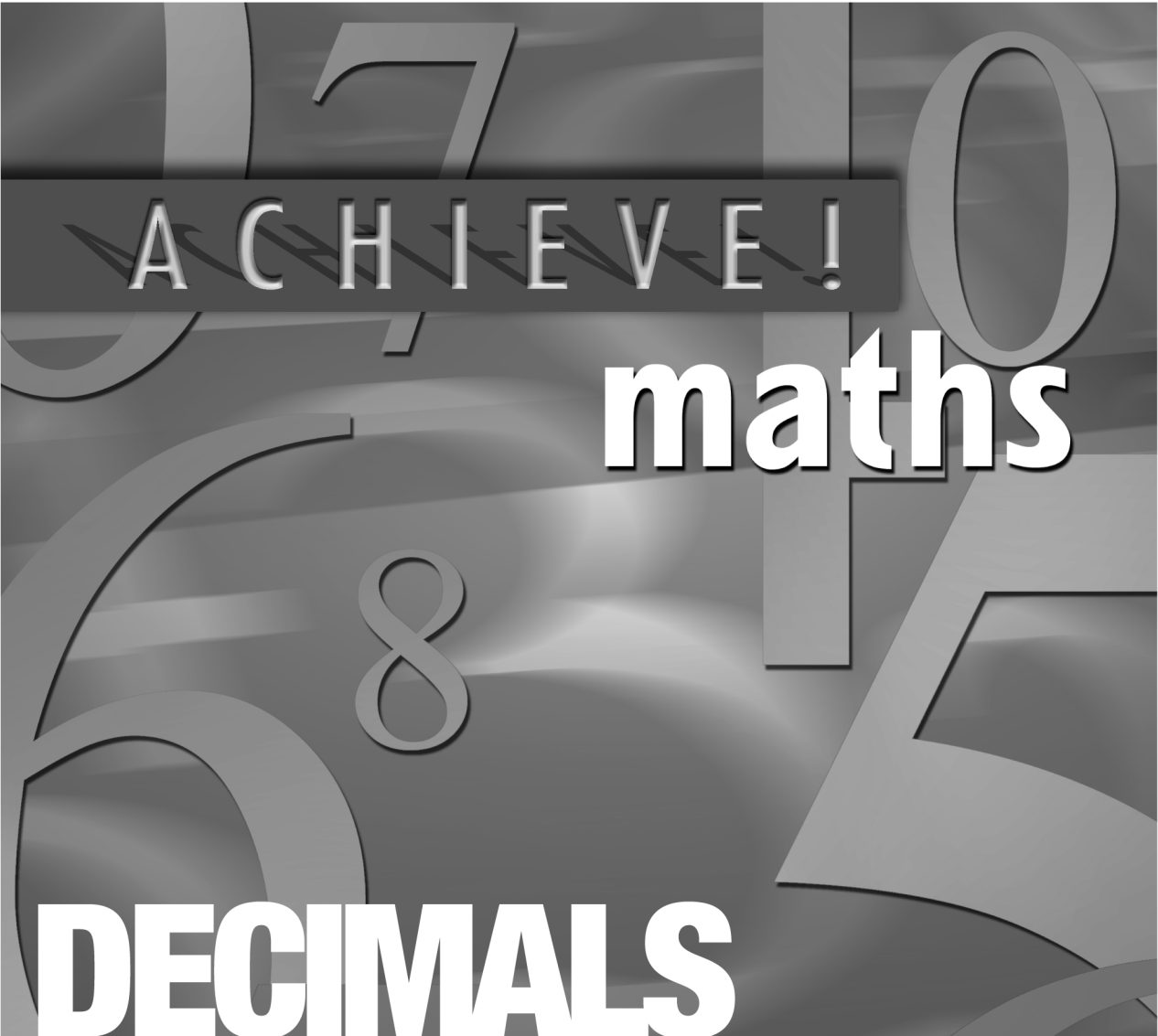


maths

**DECIMALS
& FRACTIONS**

Paul Steenson



The background of the top half of the cover is a dark grey, textured surface. It features large, stylized, 3D-rendered numbers and mathematical symbols. A large '7' is at the top left, a large '0' is at the top right, and a large '8' is in the center. The word 'ACHIEVE!' is written in a bold, sans-serif font across the top, with the letters appearing to be part of a dark, rectangular block. The word 'maths' is written in a large, white, sans-serif font with a black outline, positioned below 'ACHIEVE!'.

ACHIEVE!

maths

DECIMALS & FRACTIONS

Paul Steenson

This title was commissioned by Jennifer Gough, project management was by Piper Print Media and Gail McManus Graphics designed the cover.

Paul Steenson has more than thirty years' experience teaching Mathematics in secondary schools in Australia. He has also worked as a test development officer and been an assessor of Mathematics papers for the NSW HSC examination. Paul served as a member of the NSW Mathematics Syllabus Committee and has been a marker of both School Certificate and HSC examination papers. He has been a regular presenter of curriculum workshops and teacher development activities. He has also been a co-author of textbooks and written a number of titles in the Instant Lessons series.

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Introduction

The *Achieve!* series of Maths titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students. It is aimed at students who, because of poor literacy skills, struggle to read and comprehend regular classroom materials. Some activities are more challenging than others. The activity sheets are also suitable as supplementary material for regular classes. They may also provide useful revision in preparation for state-wide or national numeracy testing.

This book is divided into five units, each of which focuses on a topic or set of skills to do with decimals and fractions. The CD contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other activity sheets. The activity sheet documents can be easily edited to suit the needs of students and the curriculum requirements of your state.

Teachers can use the material in a variety of ways. For example, a unit could be taught in sequential lessons with a whole class or with students in groups of varying sizes. Alternatively, one or more activity sheets can be used as support material, for individual homework, assessment or revision. Teachers will need to select carefully if individual activity sheets are used as stand-alone tasks, as some activities build on knowledge gained earlier in the unit.

The Teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the activity sheets.

Prior knowledge – This refers to the skills and knowledge that are required for students to complete the tasks.

Background – This section provides a range of different types of useful information to help teach this unit. It can include explanations of key concepts and terminology or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Activity sheets and answers –

A summary of the content of each activity sheet is provided, along with teaching tips and selected answers to assist with time-efficient marking.

Extension ideas – Additional activities or discussion ideas are suggested to recap on the main skills, knowledge or terminology taught, as well as ways to extend, assess or consolidate student understanding.

Curriculum links – The activity sheets will enable students to achieve outcomes required for the following State levels or stages:

- New South Wales – Stages 4 and 5
- Victoria – Levels 5 and 6
- Queensland – Levels 4, 5 and 6



Teacher's notes

OBJECTIVES

- Write a number using words.
- Recognise and state the place value of a digit in a decimal numeral.
- Write a number in expanded notation.
- Round off a decimal to the nearest whole number.
- Round off a decimal to 1 or 2 decimal places.

PRIOR KNOWLEDGE

Students should be able to read and write whole numbers of any size. They should have a sense of the relative size of whole numbers and the role of place value in their representation.

BACKGROUND

Ensure students understand the meaning of the terms **digit**, **position** and **place value** before using these activity sheets.

When writing numbers, encourage students to leave a space between the thousands and hundreds digits in a numeral, and between the third and fourth digits after a decimal point; for example, 1 467.824 35. Commas should **not** be used instead of spaces.

Decimal is a commonly used contraction of the term **decimal fraction**. Encourage students to interpret decimals in terms of tenths, hundredths, thousandths and so on. You may need to explain the difference between place values on either side of the decimal point, such as **tens** and **tenths**, **hundreds** and **hundredths**.

Encourage students to use correct language when they read decimals. For example, the decimal 1.12 is read *one point one two* **not** *one point twelve*.

STARTER ACTIVITY

Lead a class quiz to review the terms **digit**, **position** and **place value**. For example:

- 4 tens are 40. What number is in the units place?
- Give me a number that is more than 47 and less than 6 tens.
- I'm thinking of a number. The digit in the tens position is 5, the digit in the hundreds position is 3 and the digit in the units position is 2.
- I'm thinking of a number. It has the digits 1, 2 and 3. The place value of the 1 is 10, the place value of the 2 is 200 and the place value of the 3 is 3.

Ask students to count forwards by tens or hundreds, starting on and off the decade.

ACTIVITY SHEETS

1.1 Our number system

This activity sheet uses whole numbers to remind students of the concept of place value. Read through the information at the top of the activity sheet with the students, and make sure they understand the examples of 8 736 at the top of the page and 2 549 in the first row of the table. You may need to briefly explain place value in four and five digit numbers.

Answers

1	Ten thousands	Thousands	Hundreds	Tens	Units
			3	4	1
		1	6	3	5
		4	0	7	2
1	3	8	4	6	
	4	6	2	5	
		5	0	1	
	6	7	5	0	
8	2	3	0	5	
2	0	8	1	6	
	7	0	4	0	

- 2 **a** three hundred and forty-one **b** two thousand, six hundred and thirty-five
c four hundred and six **d** one thousand, three hundred and eighty-five **e** four thousand and seventy-two **f** three thousand, six hundred and eight **g** ten thousand, five hundred and sixty-five **h** sixteen thousand, four hundred and twenty-six **i** twenty-three thousand, seven hundred and eighty **j** forty two thousand, six hundred and fifty-three

- 3 **a** 346 **b** 780 **c** 5 745 **d** 8 039 **e** 26 586
f 19 102 **g** 30 170 **h** 53 026 **i** 40 500
j 64 203 **k** 12 655 **l** 25 025

1.2 Understanding place value

This activity sheet builds on students' understanding of the place value of a digit in a number. Students are also introduced to expanded notation for whole numbers.

Answers

- 1 **a** 70 **b** 7 **c** 700 **d** 7 **e** 7 000 **f** 700
g 70 **h** 70 000
2 **a** 5 **b** 50 **c** 5 000 **d** 50 **e** 500 **f** 5 000
g 500 **h** 50 000
3 **a** $3 \times 100 + 7 \times 10 + 6 \times 1$ **b** $7 \times 100 + 5 \times 10 + 2 \times 1$ **c** $4 \times 100 + 6 \times 10$ **d** $2 \times 1\,000 + 5 \times 100 + 4 \times 10 + 8 \times 1$ **e** $5 \times 1\,000 + 7 \times 100 + 8 \times 10 + 2 \times 1$ **f** $7 \times 1\,000 + 4 \times 10 + 5 \times 1$
g $3 \times 1\,000 + 9 \times 1$ **h** $4 \times 10\,000 + 6 \times 1\,000 + 5 \times 100 + 2 \times 10 + 8 \times 1$ **i** $2 \times 10\,000 + 4 \times 100 + 8 \times 10 + 4 \times 1$ **j** $1 \times 10\,000 + 5 \times 100 + 4 \times 1$

Teacher's notes

CONTINUED

4 a 379 b 583 c 846 d 4 578 e 2 361
f 5 847 g 67 379 h 35 965 i 5 068
j 89 016 k 7 504 l 50 213

1.3 Decimal notation

This activity sheet extends the concept of place value to decimal fractions.

Answers

1	Tens	Units . Tenths	Hundredths	Thousandths
		7 . 6	5	
1		4 . 8	2	
		6 . 3	4	5
3		5 . 6		
5		7 . 5	2	6
		9 . 0	5	
4		0 . 3		
2		3 . 0	7	5
		0 . 0	9	5
		8 . 0	0	3

2 a six units and three tenths b eight units and five tenths c one ten, four units, two tenths and eight hundredths d three tens, two units, four tenths and six hundredths e five units, two tenths, eight hundredths and six thousandths f one ten, five units, seven tenths, two hundredths and three thousandths g two tens, nine units and two hundredths h seven units, nine hundredths and five thousandths i four units, one tenth and five thousandths j three hundreds, four units and six tenths k five tens, six units and seven hundredths l eight tens, four units, three tenths, seven hundredths and five thousandths

3 a 6.7 b 4.23 c 35.8 d 24.09 e 50.62
f 3.684 g 0.895 h 0.073

1.4 Decimal place value

This activity sheet continues to develop students' understanding of place value in decimal numbers. It extends the work in the previous activity sheet, 1.3 Decimal notation.

Answers

1 a $4 \times 1 + 3 \times \frac{1}{10}$ b $3 \times 1 + 2 \times \frac{1}{10}$
c $4 \times 1 + 6 \times \frac{1}{10} + 2 \times \frac{1}{100}$
d $6 \times 1 + 1 \times \frac{1}{10} + 9 \times \frac{1}{100}$ e $2 \times 1 + 4 \times \frac{1}{100}$
f $4 \times \frac{1}{10} + 9 \times \frac{1}{100} + 3 \times \frac{1}{1000}$
g $7 \times 1 + 6 \times \frac{1}{10} + 2 \times \frac{1}{100} + 9 \times \frac{1}{1000}$
h $2 \times 10 + 4 \times 1 + 1 \times \frac{1}{10} + 6 \times \frac{1}{100} + 8 \times \frac{1}{1000}$
2 a 6.3 b 5.8 c 7.56 d 3.735 e 6.94
f 92.79 g 20.5 h 8.046

3 a 3 tenths b 3 tenths c 3 hundredths
d 3 tenths e 3 thousandths f 3 hundredths
g 3 hundredths h 3 tenths
4 a 8 tenths b 8 hundredths c 8 units
d 8 tenths e 8 hundredths f 8 thousandths
g 8 tenths h 8 hundredths

1.5 Rounding off decimals A

This activity sheet first develops the skill of rounding decimal numbers to the nearest whole number. Students then learn to round numbers correct to 1 or 2 decimal places. This activity sheet could be used independently or when students are working with the multiplication or division of decimal numbers.

Answers

1 a 4 b 6 c 2 d 4 e 7 f 5 g 6 h 6
i 8 j 4 k 10 l 12 m 14 n 17 o 16
p 8 q 6 r 19
2 a 9.3 b 7.7 c 5.2 d 6.2 e 8.1 f 4.5
g 9.3 h 6.5 i 13.6 j 10.2 k 12.6 l 23.5
m 0.5 n 2.0 o 20.0
3 a 7.92 b 9.24 c 3.78 d 6.93 e 0.16
f 17.42 g 20.41 h 10.30 i 13.04 j 17.00
k 9.14 l 8.62 m 15.78 n 22.75 o 12.10

1.6 Rounding off decimals B

This activity sheet provides students with practice in rounding decimals arising from calculations in measurement and money.

Answers

1 a 7 m b 5 kg c 5 L d 5 m e 9 cm
f 7 kg g 21 km h 21 kg i 15 L j 14 m
k 9 m l 13 L m 16 kg n 28 L o 33 km
2 a 6.3 m b 8.7 kg c 9.5 L d 7.6 kg
e 3.5 m f 21.3 km g 28.6 kg h 40.4 m
i 18.8 cm j 33.0 L k 7.5 kg l 4.1 m
m 15.6 L n 26.3 km o 45.4 kg
3 a \$4.29 b \$9.67 c \$12.45 d \$10.07
e \$14.74 f \$321.33 g \$114.12 h \$27.76
i \$734.65 j \$25.35 k \$18.54 l \$7.64

EXTENSION IDEAS

Write 4 digits on the board, such as 2 3 4 5. Have students work in pairs to write the three largest numbers they can using these digits, and the three smallest numbers they can. Encourage students to describe the digits in the numbers they write in terms of place value, such as 6 thousand or 2 hundredths. This activity can also be done using 5 or 6 digits.

1.1 Our number system

The number system we use today was devised by the Hindu people who lived in the area of modern India. It was taken to Europe by Arab traders.

Our number system uses ten symbols called **digits**: 1, 2, 3, 4, 5, 6, 7, 8, 9 and 0. The position of each digit determines its **place value** in a numeral.

For example, **8 736** means **8** thousands, **7** hundreds, **3** tens and **6** units.

1 Complete this place value table. The first one is done for you.

	Ten thousands	Thousands	Hundreds	Tens	Units
2 549 represents		2	5	4	9
341 represents					
1 635 represents					
4 072 represents					
13 846 represents					
4 625 represents					
501 represents					
6 750 represents					
82 305 represents					
20 816 represents					
7 040 represents					

2 Write each number in words.

EXAMPLE

235 is two hundred and thirty-five



- a 341 _____
- b 2 635 _____
- c 406 _____

1.1 Our number system

CONTINUED

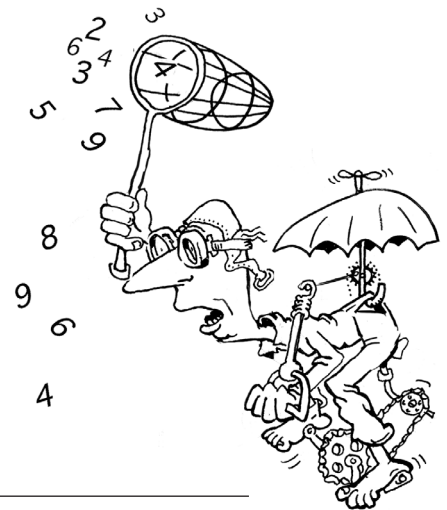
- d 1 385 _____
- e 4 072 _____
- f 3 608 _____
- g 10 565 _____
- h 16 426 _____
- i 23 780 _____
- j 42 653 _____

3 Write the numeral for each of the following.

EXAMPLE

six hundred and seventy-two = 672

- a three hundred and forty-six _____
- b seven hundred and eighty _____
- c five thousand, seven hundred and forty-five _____
- d eight thousand and thirty-nine _____
- e twenty-six thousand, five hundred and eighty-six _____
- f nineteen thousand, one hundred and two _____
- g thirty thousand, one hundred and seventy _____
- h fifty-three thousand and twenty-six _____
- i forty thousand, five hundred _____
- j sixty-four thousand, two hundred and three _____
- k twelve thousand, six hundred and fifty-five _____
- l twenty-five thousand and twenty-five _____



1.2 Understanding place value

The **position** of each digit determines its **value** in a numeral.

In the number 647, the place value of the 6 is 600, the place value of the 4 is 40 and the place value of the 7 is simply 7.

647 can be written as $600 + 40 + 7$.

Each digit position is 10 times the value of the digit position to its right.

1 What is the place value of the digit 7 in each numeral?

a 873 70

b 37

c 765

d 367

e 7 458

f 2 716

g 9 070

h 73 490

2 What is the place value of the digit 5 in each numeral?

a 95

b 753

c 5 416

d 1 250

e 8 567

f 15 432

g 87 549

h 56 743

One way to show the place value of each digit in a number is to use **expanded notation**.

EXAMPLE

Write 5 369 in expanded notation.

$$5\,369 = 5 \times 1\,000 + 3 \times 100 + 6 \times 10 + 9 \times 1$$



3 Write these numbers in expanded notation.

a 376

b 752

c 460

d 2 548

1.2 Understanding place value

CONTINUED

e 5 782 _____

f 7 045 _____

g 3 009 _____

h 46 528 _____

i 20 484 _____

j 10 504 _____

4 Write the basic numeral for each of these numbers.

EXAMPLE

$$3 \times 1\,000 + 4 \times 100 + 5 \times 10 + 7 \times 1 = \underline{3\,457}$$

a $3 \times 100 + 7 \times 10 + 9 \times 1$ _____b $5 \times 100 + 8 \times 10 + 3 \times 1$ _____c $8 \times 100 + 4 \times 10 + 6 \times 1$ _____d $4 \times 1\,000 + 5 \times 100 + 7 \times 10 + 8 \times 1$ _____e $2 \times 1\,000 + 3 \times 100 + 6 \times 10 + 1 \times 1$ _____f $5 \times 1\,000 + 8 \times 100 + 4 \times 10 + 7 \times 1$ _____g $6 \times 10\,000 + 7 \times 1\,000 + 3 \times 100 + 7 \times 10 + 9 \times 1$ _____h $3 \times 10\,000 + 5 \times 1\,000 + 9 \times 100 + 6 \times 10 + 5 \times 1$ _____i $5 \times 1\,000 + 6 \times 10 + 8 \times 1$ _____j $8 \times 10\,000 + 9 \times 1\,000 + 1 \times 10 + 6 \times 1$ _____k $7 \times 1\,000 + 5 \times 100 + 4 \times 1$ _____l $5 \times 10\,000 + 2 \times 100 + 1 \times 10 + 3 \times 1$ _____

1.3 Decimal notation

People use decimals every day. When you write an amount of money, \$3.25, or a measurement, 6.9 metres, you are using decimals to write whole numbers and parts of that unit.

The **position** of a digit determines its **place value**. The digits after the decimal point indicate **fractions** (parts) of the unit such as tenths, hundredths and thousandths.

3.25 means 3 units, 2 tenths and 5 hundredths

Or: $3 \times 1 + 2 \times \frac{1}{10} + 5 \times \frac{1}{100}$



1 Complete this place value table. The first one is done for you.

	Tens	Units	.	Tenths	Hundredths	Thousandths
8.42 represents		8	.	4	2	
7.65 represents			.			
14.82 represents			.			
6.345 represents			.			
35.6 represents			.			
57.526 represents			.			
9.05 represents			.			
40.3 represents			.			
23.075 represents			.			
0.095 represents			.			
8.003 represents			.			

2 Write each number in words.

EXAMPLE

5.63 is five units, six tenths and three hundredths

a 6.3 _____

b 8.5 _____

1.3 Decimal notation

CONTINUED

- c 14.28 _____
- d 32.46 _____
- e 5.286 _____
- f 15.723 _____
- g 29.02 _____
- h 7.095 _____
- i 4.105 _____
- j 304.6 _____
- k 56.07 _____
- l 84.375 _____

3 Write each of the following as a decimal numeral.

EXAMPLE

three units, two tenths and six hundredths = 3.26

- a six units and seven tenths _____
- b four units, two tenths and three hundredths _____
- c three tens, five units and eight tenths _____
- d two tens, four units and nine hundredths _____
- e five tens, six tenths and two hundredths _____
- f three units, six tenths, eight hundredths and four thousandths _____
- g eight tenths, nine hundredths and five thousandths _____
- h seven hundredths and three thousandths _____

1.4 Decimal place value

The **position** of each digit determines its **value** in a decimal numeral.

A number such as 3.925 can be written in different ways.

$$\begin{aligned} 3.925 &= 3 + 0.9 + 0.02 + 0.005 \\ &= 3 \text{ units} + 9 \text{ tenths} + 2 \text{ hundredths} + 5 \text{ thousandths} \\ &= 3 \times 1 + 9 \times \frac{1}{10} + 2 \times \frac{1}{100} + 5 \times \frac{1}{1000} \end{aligned}$$

1 Write these decimals in expanded notation.

EXAMPLE

$$4.67 = 4 \times 1 + 6 \times \frac{1}{10} + 7 \times \frac{1}{100}$$



a 4.3 _____

b 3.2 _____

c 4.62 _____

d 6.19 _____

e 2.04 _____

f 0.493 _____

g 7.629 _____

h 24.168 _____

2 Write in decimal form.

EXAMPLE

$$4 \times 1 + 8 \times \frac{1}{10} + 5 \times \frac{1}{100} = \underline{4.85}$$

a $6 \times 1 + 3 \times \frac{1}{10}$ _____

b $5 \times 1 + 8 \times \frac{1}{10}$ _____

c $7 \times 1 + 5 \times \frac{1}{10} + 6 \times \frac{1}{100}$ _____

1.4 Decimal place value

CONTINUED

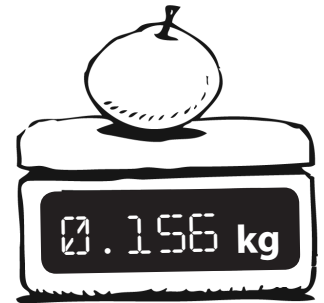
d $3 \times 1 + 7 \times \frac{1}{10} + 3 \times \frac{1}{100} + 5 \times \frac{1}{1000}$ _____

e $6 \times 1 + 9 \times \frac{1}{10} + 4 \times \frac{1}{100}$ _____

f $9 \times 10 + 2 \times 1 + 7 \times \frac{1}{10} + 9 \times \frac{1}{100}$ _____

g $2 \times 10 + 5 \times \frac{1}{10}$ _____

h $8 \times 1 + 4 \times \frac{1}{100} + 6 \times \frac{1}{1000}$ _____



EXAMPLE

In the number **0.576**, the place value of the digit **5** is 5 tenths ($\frac{5}{10}$), the place value of the **7** is 7 hundredths ($\frac{7}{100}$) and the place value of the **6** is 6 thousandths ($\frac{6}{1000}$).

3 What is the place value of the digit **3** in each of these decimal numerals?

a 0.3 _____ **3 tenths**

b 29.3 _____

c 8.53 _____

d 4.38 _____

e 8.473 _____

f 5.03 _____

g 14.735 _____

h 26.302 _____

4 What is the place value of the digit **8** in each of these numerals?

a 4.8 _____

b 1.08 _____

c 8.6 _____

d 12.87 _____

e 23.68 _____

f 2.318 _____

g 7.843 _____

h 19.085 _____

1.5 Rounding off decimals A

Josie's calculator showed an answer as:

15.63972

The question required an answer to the nearest whole number.

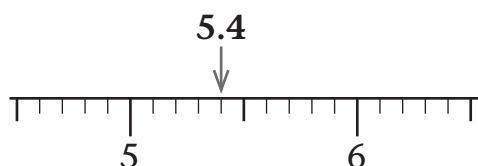
When using decimals, we often have more digits after the decimal point than we need. This may happen after completing a multiplication or division.

If this does happen, we **round off** to the number of decimal places that we need and leave out the following digits. Looking at a number line or the scale on your ruler can make this easier.

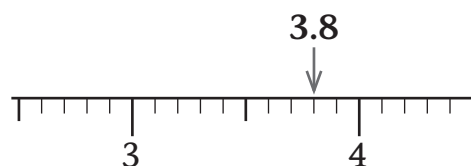


EXAMPLES

5.4 is closer to 5 than to 6, so 5.4 rounded to the nearest whole number gives 5.



3.8 is closer to 4 than to 3, so 3.8 rounded to the nearest whole number gives 4.



Rule for rounding a decimal

- 1 Cut the numeral at the required place.
- 2 Look at the next digit to the right of the cut.
- 3 If this next digit is 0, 1, 2, 3 or 4, leave the previous digit unchanged. (This is called **rounding down**.)

If this next digit is 5, 6, 7, 8 or 9, add 1 to the previous digit. (This is called **rounding up**.)

EXAMPLE

Round 15.7 to the nearest whole number.

15.7
cut

The next digit after the cut is 7.

Applying the rule, we round the previous digit **up**.

The answer is **16** to the nearest whole number.

- 1 Round these decimals to the nearest whole number.
You can use the scale on your ruler or a number line to help.

a 4.3 _____

b 5.7 _____

c 2.4 _____

1.5 Rounding off decimals A

CONTINUED

d 3.9 _____

e 7.2 _____

f 4.6 _____

g 6.3 _____

h 5.8 _____

i 8.1 _____

j 3.5 _____

k 10.2 _____

l 12.4 _____

m 13.7 _____

n 16.5 _____

o 15.8 _____

p 8.16 _____

q 6.08 _____

r 19.45 _____

EXAMPLE

Round 8.263 5 to 1 decimal place.

8.263 5
cut

The next digit after the cut is 6.

Applying the rule, we round the previous digit **up**.The answer is **8.3** correct to 1 decimal place.

- 2** Write these decimals correct to 1 decimal place.
(Round to the nearest tenth.)

a 9.31 _____

b 7.66 _____

c 5.24 _____

d 6.15 _____

e 8.07 _____

f 4.53 _____

g 9.347 _____

h 6.499 _____

i 13.613 _____

j 10.173 _____

k 12.625 _____

l 23.492 _____

m 0.505 _____

n 2.037 _____

o 19.95 _____

- 3** Write these decimals correct to 2 decimal places.
(Round to the nearest hundredth.)

a 7.923 _____

b 9.238 _____

c 3.784 _____

d 6.929 _____

e 0.158 _____

f 17.418 _____

g 20.405 _____

h 10.298 _____

i 13.042 _____

j 16.996 _____

k 9.136 4 _____

l 8.624 7 _____

m 15.781 5 _____

n 22.752 8 _____

o 12.097 4 _____

1.6 Rounding off decimals B

Rule for rounding a decimal

- 1 Cut the numeral at the required place.
- 2 Look at the next digit to the right of the cut.
- 3 If this next digit is 0, 1, 2, 3 or 4, leave the previous digit unchanged (This is called **rounding down**.)
If this next digit is 5, 6, 7, 8 or 9, add 1 to the previous digit (This is called **rounding up**.)

1 Round these measurements to the nearest **whole unit** (whole number).

- | | | |
|-------------------------|------------------------|-------------------------|
| a 6.8 m _____ | b 4.9 kg _____ | c 5.3 L _____ |
| d 4.7 m _____ | e 8.5 cm _____ | f 7.2 kg _____ |
| g 21.4 km _____ | h 20.6 kg _____ | i 15.1 L _____ |
| j 14.3 m _____ | k 9.37 m _____ | l 12.56 L _____ |
| m 16.09 kg _____ | n 28.34 L _____ | o 32.53 km _____ |

2 Round these measurements correct to **1 decimal place**.

- | | | |
|-------------------------|--------------------------|--------------------------|
| a 6.34 m _____ | b 8.67 kg _____ | c 9.46 L _____ |
| d 7.58 kg _____ | e 3.52 m _____ | f 21.25 km _____ |
| g 28.59 kg _____ | h 40.35 m _____ | i 18.77 cm _____ |
| j 33.03 L _____ | k 7.481 kg _____ | l 4.095 m _____ |
| m 15.605 L _____ | n 26.345 km _____ | o 45.382 kg _____ |

3 Write the following amounts of money to the **nearest cent**. (Correct to 2 decimal places.)

- | | |
|----------------------------|----------------------------|
| a \$4.293 _____ | b \$9.666 _____ |
| c \$12.451 _____ | d \$10.069 _____ |
| e \$14.736 _____ | f \$321.333 3 _____ |
| g \$114.124 5 _____ | h \$27.757 5 _____ |
| i \$734.654 _____ | j \$25.345 5 _____ |
| k \$18.535 3 _____ | l \$7.639 02 _____ |

Teacher's notes

OBJECTIVES

- Add and subtract decimal numbers.
- Add and subtract amounts of money or measurements.
- Solve simple word problems involving addition or subtraction of decimal numbers.

PRIOR KNOWLEDGE

Students should be able to write, read and interpret numbers written in decimal notation. They should be able to add and subtract whole numbers of any size.

BACKGROUND

Money and measurements are common applications of decimal numbers. Many students will be able to add and subtract amounts of money. This can be a useful starting point for developing calculation skills with decimals.

These activity sheets have been developed for students to complete without using calculators. If students are using calculators, they must learn to interpret the display in the context of the problem; for example, 2.6 means \$2.60 when using money.

You could have students complete some activity sheets using pen and paper methods and then use calculators to check their answers. This will reinforce both pen and paper and calculator skills.

STARTER ACTIVITY

Use the context of money to introduce the concept of decimal fractions. Ask questions such as, 'How many 50 cent pieces are equal to one dollar?' (2) 'How do we write 50 cents as part of one dollar?' (\$0.50). Extend this idea to the number of 20 cent pieces in one dollar, and 10 and 5 cent pieces.

Encourage students to distinguish between the language for describing decimals and for describing amounts of money. For example, \$1.50 is said 'one dollar fifty' whereas the fraction 1.5 is said 'one point five'.

ACTIVITY SHEETS

2.1 Adding decimals A

This activity sheet develops students' skills in adding decimal numbers without the use of a calculator. The questions are set out in columns to familiarise students with this important step when using pen and paper methods. Encourage students to use zeros to fill any gaps to the right of the decimal point to equalise the number of decimal places; for example, when adding 4.25 and 3.7, include a place holder 0 by writing 3.70.

Answers

1 a 3.9 b 3.4 c 4.2 d 6.7 e 3.68 f 4.51
g 6.58 h 9.11 i 7.95 j 12.73 k 41.75
l 18.42 m 22.22 n 40.78 o 39.52
p 32.45

2.2 Adding decimals B

This activity sheet extends students' addition skills with decimal numbers. Encourage students to set their work out in columns as shown in the example. In the word problems (question 3) it is suggested that you help students identify the key words which indicate an addition is required and ask them to underline these words. For the questions on the second page, students need to write their working and answers in their workbooks.

Answers

1 a 24.55 b 8.99 c 12.38 d 107.12
e 19.21 f 82.48 g 50.23
2 a 72.24 b \$21.51 c \$11.97 d \$50.90
e 37 km f 39.3 m g 33.81 kg h 26.48 L
i 67.5 kg
3 a \$16.40 b \$21.34 c \$24.49 d \$22.70
e 9.8 m f 26.4 kg g \$37.99
4 \$5.84
5 10.5 m

Teacher's notes

CONTINUED

2.3 Subtracting decimals A

This activity sheet develops students' skills in subtracting decimal numbers without the use of a calculator. The questions are set out in columns to familiarise students with this important step when using pen and paper methods. Encourage students to use zeros to fill any gaps to the right of the decimal point to equalise the number of decimal places.

Answers

1 a 1.3 b 0.5 c 1.4 d 2.6 e 1.45 f 2.13
 g 0.52 h 2.54 i 11.1 j 7.25 k 5.37
 l 13.26 m 2.54 n 11.63 o 5.99 p 9.17
 q 11.35 r 26.31 s 1.13 t 2.95

2.4 Subtracting decimals B

This activity sheet extends students' subtraction skills with decimal numbers. Encourage students to set their work out in columns as shown in the example. In the word problems (question 3) it is suggested that you help students identify the key words which indicate a subtraction is required and ask them to underline these words. For the questions on the second page, students need to write their working and answers in their workbooks.

Answers

1 a 0.5 b 0.4 c 2.2 d 7.14 e 2.61 f 3.43
 g 1.47 h 4.47 i 13.96 j 2.61
 2 a \$8.90 b \$3.91 c \$9.79 d \$37.70
 e 3.6 kg f 49.8 cm g 21.7 m h 28.1 L
 i 33.08 kg j 28.01 km
 3 a \$1.65 b \$5.35 c \$3.32 d \$4.95
 e \$18.45 f \$9.26
 4 7.8 m
 5 1.63 m
 6 20.88 m
 7 32.4 L

EXTENSION IDEAS

Have the students practise their mental maths skills at 'making units'. For example, 'How much do you add to 0.45 to make 1?'

Have students check their solutions by using the inverse operation. For example, check the subtraction $15.7 - 7.95 = 7.75$ by adding the answer and the number that was subtracted ($7.75 + 7.95 = 15.7$).

Give students a copy of the weekly supermarket specials and a shopping list. Ask them to find the price of each item on the list and then calculate the total cost of the shopping list.

To extend decimal calculation skills further, you could provide a variety of store catalogues, and designate a value such as \$50. Students work in pairs, with one student choosing an item to buy, and using a calculator to complete the subtraction to find out how much money is left over, while the second student uses the pen and paper method to complete the same calculation.

2.1 Adding decimals A

Kayla measured the amount of rainwater collected in three containers: 11.36 litres, 22.5 litres and 7.63 litres.

She added these numbers to calculate the total amount of water that had been collected.

Careful setting out is important when adding decimal numbers.

When you add decimal numbers:

- write the numbers under one another with the digits in their correct place value columns
- keep the decimal points under each other.



EXAMPLE

$$11.36 + 22.5 + 7.63$$

It may help to use zeros as place holders, to fill any gaps to the right of the decimal point.

$$\begin{array}{r} 11.36 \\ + 22.50 \\ + 7.63 \\ \hline 41.49 \end{array}$$

place
holder
0

1 Complete these addition calculations.

a

$$\begin{array}{r} 2.6 \\ + 1.3 \\ \hline \end{array}$$

b

$$\begin{array}{r} 1.8 \\ + 1.6 \\ \hline \end{array}$$

c

$$\begin{array}{r} 2.3 \\ + 1.9 \\ \hline \end{array}$$

d

$$\begin{array}{r} 4.2 \\ + 2.5 \\ \hline \end{array}$$

e

$$\begin{array}{r} 2.25 \\ + 1.43 \\ \hline \end{array}$$

f

$$\begin{array}{r} 3.15 \\ + 1.36 \\ \hline \end{array}$$

g

$$\begin{array}{r} 3.75 \\ + 2.83 \\ \hline \end{array}$$

h

$$\begin{array}{r} 5.26 \\ + 3.85 \\ \hline \end{array}$$

i

$$\begin{array}{r} 4.25 \\ + 3.7 \\ \hline \end{array}$$

j

$$\begin{array}{r} 5.83 \\ + 6.9 \\ \hline \end{array}$$

k

$$\begin{array}{r} 28.3 \\ + 13.45 \\ \hline \end{array}$$

l

$$\begin{array}{r} 8.5 \\ + 9.92 \\ \hline \end{array}$$

m

$$\begin{array}{r} 13.87 \\ + 3.28 \\ + 5.07 \\ \hline \end{array}$$

n

$$\begin{array}{r} 8.73 \\ + 3.7 \\ + 28.35 \\ \hline \end{array}$$

o

$$\begin{array}{r} 15.7 \\ + 9.65 \\ + 3.9 \\ \hline 10.27 \end{array}$$

p

$$\begin{array}{r} 3.17 \\ + 2.5 \\ + 15.08 \\ \hline 11.7 \end{array}$$

2.2 Adding decimals B

Jon bought a chocolate bar for \$3.89 and a large drink for \$5.25.
What is the total cost? To obtain the answer, we need to add these decimals.

EXAMPLE

$$3.89 + 5.25$$

When you add decimal numbers:

- write the numbers under one another with the digits in their correct place value columns
- keep the decimal points under each other.

$$\begin{array}{r} 3.89 \\ + 5.25 \\ \hline 9.14 \end{array}$$

The total cost is **\$9.14**.

1 Complete these addition calculations in the spaces provided.

a $16.2 + 8.35$

b $3.24 + 1.95 + 3.8$

c $3.9 + 6.58 + 1.9$

d $26.3 + 51.42 + 29.4$

e $1.37 + 6.7 + 9.42 + 1.72$

f $31.46 + 24.9 + 19.32 + 6.8$

g $16.3 + 17.65 + 8.38 + 7.9$

2.2 Adding decimals B

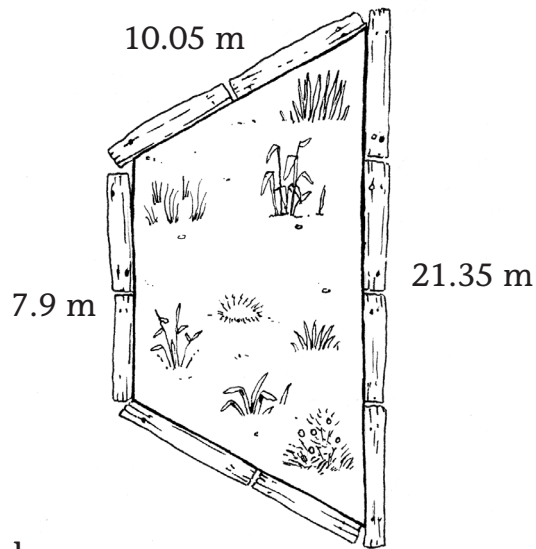
CONTINUED

Answer the following questions in your workbook.

Remember to set your work out carefully and keep the decimal points under each other.

2 Add.

- a** $15.5 + 13.75 + 27.82 + 15.17$
- b** $\$12.67 + \8.84
- c** $\$6.18 + \$3.29 + \$2.50$
- d** $\$25.50 + \$14.95 + \$10.45$
- e** $16.3 \text{ km} + 8.5 \text{ km} + 12.2 \text{ km}$
- f** $7.9 \text{ m} + 21.35 \text{ m} + 10.05 \text{ m}$
- g** $9.42 \text{ kg} + 12.49 \text{ kg} + 11.9 \text{ kg}$
- h** $6.25 \text{ L} + 7.73 \text{ L} + 12.5 \text{ L}$
- i** $16.8 \text{ kg} + 7.95 \text{ kg} + 23.4 \text{ kg} + 19.35 \text{ kg}$



3 Complete these additions.

- a** Add $\$9.45$ and $\$6.95$.
- b** What is the sum of $\$12.85$ and $\$8.49$?
- c** What is $\$8.50$ more than $\$15.99$?
- d** Increase $\$16.75$ by $\$5.95$.
- e** Add 4.3 m , 1.9 m and 3.6 m .
- f** What is the sum of 14.8 kg and 11.6 kg ?
- g** What is the total of $\$13.46$, $\$14.55$ and $\$9.98$?



- 4** Katie purchased a ruler ($\$1.85$) and a set of coloured pencils ($\$3.99$). How much did she spend?
- 5** The 3 sides of a triangle measure 4.7 m , 3.2 m and 2.6 m . Add these lengths to calculate the perimeter of this triangle.

2.3 Subtracting decimals A

Toni had \$15.70 and spent \$7.95. She subtracted these amounts to check how much money she should still have in her pocket.

Careful setting out is important when subtracting decimal numbers.

When you subtract decimal numbers:

- write the numbers under one another with the digits in their correct place value columns
- keep the decimal points under each other
- use zeros as place holders to fill any gaps to the right of the decimal point.

EXAMPLE

$$\begin{array}{r} 15.70 - 7.95 \\ \underline{7.95} \\ 7.75 \end{array}$$

1 Complete these subtraction calculations.

a
$$\begin{array}{r} 2.5 - 1.2 \\ \underline{} \end{array}$$

b
$$\begin{array}{r} 1.9 - 1.4 \\ \underline{} \end{array}$$

c
$$\begin{array}{r} 4.3 - 2.9 \\ \underline{} \end{array}$$

d
$$\begin{array}{r} 5.4 - 2.8 \\ \underline{} \end{array}$$

e
$$\begin{array}{r} 2.86 - 1.41 \\ \underline{} \end{array}$$

f
$$\begin{array}{r} 3.27 - 1.14 \\ \underline{} \end{array}$$

g
$$\begin{array}{r} 2.39 - 1.87 \\ \underline{} \end{array}$$

h
$$\begin{array}{r} 7.32 - 4.78 \\ \underline{} \end{array}$$

i
$$\begin{array}{r} 23.7 - 12.6 \\ \underline{} \end{array}$$

j
$$\begin{array}{r} 19.74 - 12.49 \\ \underline{} \end{array}$$

k
$$\begin{array}{r} 7.34 - 1.97 \\ \underline{} \end{array}$$

l
$$\begin{array}{r} 21.04 - 7.78 \\ \underline{} \end{array}$$

m
$$\begin{array}{r} 21.08 - 18.54 \\ \underline{} \end{array}$$

n
$$\begin{array}{r} 30.57 - 18.94 \\ \underline{} \end{array}$$

o
$$\begin{array}{r} 15.79 - 9.8 \\ \underline{} \end{array}$$

p
$$\begin{array}{r} 18.9 - 9.73 \\ \underline{} \end{array}$$

q
$$\begin{array}{r} 27.75 - 16.4 \\ \underline{} \end{array}$$

r
$$\begin{array}{r} 43.6 - 17.29 \\ \underline{} \end{array}$$

s
$$\begin{array}{r} 2.73 - 1.6 \\ \underline{} \end{array}$$

t
$$\begin{array}{r} 9.45 - 6.5 \\ \underline{} \end{array}$$

2.4 Subtracting decimals B

Tuyen cut 1.3 m from a length of wood 2.5 m long.
How long is the remaining piece of wood?
To obtain the answer, we need to subtract 1.3 from 2.5.



EXAMPLE

$2.5 - 1.3$

When you subtract decimal numbers:

- write the numbers under one another with the digits in their correct place value columns
- write the larger number first (on the top line)
- keep the decimal points under each other.

$$\begin{array}{r} 2.5 \\ - 1.3 \\ \hline 1.2 \end{array}$$

1.2 m remains.

1 Complete these subtraction calculations in the spaces provided.

a $0.7 - 0.2$

f $7.82 - 4.39$

b $2.3 - 1.9$

g $2.7 - 1.23$

c $5.8 - 3.6$

h $6.4 - 1.93$

d $11.43 - 4.29$

i $23.76 - 9.8$

e $9.02 - 6.41$

j $10 - 7.39$

2.4 Subtracting decimals B

CONTINUED

Answer the following questions in your workbook.

Remember to set your work out carefully and keep the decimal points under each other.

2 Subtract.

a $\$19.73 - \10.83

b $\$9.56 - \5.65

c $\$29.35 - \19.56

d $\$107 - \69.30

e $19.2 \text{ kg} - 15.6 \text{ kg}$

f $95.7 \text{ cm} - 45.9 \text{ cm}$

g $120.3 \text{ m} - 98.6 \text{ m}$

h $56 \text{ L} - 27.9 \text{ L}$

i $132.48 \text{ kg} - 99.4 \text{ kg}$

j $34.3 \text{ km} - 6.29 \text{ km}$

3 Complete these subtractions.

a Subtract $\$8.35$ from $\$10$.

b What is $\$20$ minus $\$14.65$?

c Take $\$11.68$ away from $\$15$.

d What amount is $\$5$ less than $\$9.95$?

e Decrease $\$26.80$ by $\$8.35$.

f What is the difference between $\$28.95$ and $\$19.69$?

4 Kevin cut 7.9 metres of material from a length of 15.7 metres.
What length of material is left?

5 A netball ring is 3.05 m above the ground.
Erin is 1.42 m tall.
What is the difference between these heights?

6 Kaitlin's best discus throw was 52.73 m and
her best javelin throw was 31.85 m .
What is the difference between these distances?

7 The petrol tank of a car holds 60 litres when full.
After a week's travel, 27.6 litres remain.
How many litres are needed to fill the tank?



Teacher's notes

OBJECTIVES

- Multiply a decimal number by another decimal number.
- Divide a decimal number by a whole number.
- Divide a decimal number by another decimal.
- Solve simple word problems involving multiplication or division of decimal numbers.
- Solve simple word problems involving money.

PRIOR KNOWLEDGE

Students should be able to write, read and interpret numbers written in decimal notation. They should be able to multiply and divide whole numbers by 1 and 2 digit numbers.

BACKGROUND

Money and measurements are common applications of decimal numbers in everyday life. This can be a useful starting point for developing calculation skills with decimals.

These activity sheets have been developed for students to complete without using calculators.

You could have students complete some activity sheets using pen and paper methods and then use calculators to check their answers. This will reinforce both pen and paper and calculator skills.

STARTER ACTIVITY

Challenge students to think of real-life situations where people need to multiply or divide by decimals. List their ideas on the board, and include sample calculations. If they need hints, suggest:

- working out costs of buying products by the kilogram or by the length
- calculating area
- dividing amounts into equal parts, such as money or two pizzas
- financial contexts such as working out hourly rates of pay, interest payable or tax payable.

ACTIVITY SHEETS

3.1 Multiplying decimals A

This activity sheet develops students' skills in multiplying decimal numbers without the use of a calculator. The questions are limited to multiplication by a single digit number.

Answers

- 1 a 0.14 b 0.54 c 0.32 d 0.01 e 0.028
f 0.004 5 g 0.46 h 2.52 i 2.36 j 3.15
k 2.65 l 0.206 m 0.432 n 0.072 o 1.172
p 2.7 q 1.96 r 0.815 s 0.893 2 t 7.468
u 1.932
2 a \$68.75 b \$37.40 c \$29.12 d \$84.60

3.2 Multiplying decimals B

This activity sheet extends students' multiplication skills with decimal numbers. Students need to be able to multiply by two digit numbers to complete the tasks. For this activity sheet, students need to write their working and answers in their workbooks.

Answers

- 1 a 4.14 b 5.22 c 9.996 d 2.859 5
e 525.42 f 59.312 5 g 6.363 5 h 135.45
i 1.232 4 j 48.246 k 262.5 l 416.34
2 a 14.62 m b 13.32 kg c 36.57 L d \$10.08
e 140.75 kg f \$18.30 g 91.28 m h 72.54L
i \$208.89
3 \$4.47
4 4.68 m²
5 \$17
6 \$267.33

3.3 Dividing decimals by whole numbers A

This activity sheet develops students' skills in dividing a decimal by a whole number.

Answers

- 1 a 3.4 b 3.2 c 3.1 d 6.2 e 4.2 f 4.2
g 6.6 h 1.57 i 1.62 j 6.31 k 2.56
l 3.12 m 0.314 n 0.125 o 0.126 p 0.007
q 0.106 r 0.067 s 0.091 t 1.175 u 2.32
v 1.035 w 3.426 x 2.195
2 a 2.1 b 1.3 c 3.8 d 2.6 e 4.3 f 1.14
g 1.48 h 1.23 i 0.09 j 0.08 k 0.65
l 0.89
3 a 1.23 b 0.65 c 0.59 d 1.13 e 13.77
f 6.18 g 2.69 h 7.31 i 3.49 j 3.76
k 4.06 l 1.42

Teacher's notes

CONTINUED

3.4 Dividing decimals by whole numbers B

This activity sheet includes questions involving money and measurements and a selection of word problems. For this activity sheet, students need to write their working and answers in their workbooks.

Answers

- 1 **a** \$1.70 **b** \$7.30 **c** \$42.40 **d** 0.62 kg
 e 5.6 L **f** \$3.92 **g** 2.15 m **h** 1.84 kg
 i 36.1 m
- 2 **a** \$2.98 **b** \$1.79 **c** \$1.96 **d** \$2.93
 e \$8.90 **f** \$8.99 **g** \$4.72 **h** \$6.96 **i** \$8.43
- 3 0.85m
- 4 \$1.75
- 5 \$1.85
- 6 \$11.55

3.5 Dividing by a decimal A

This activity sheet extends students' division skills to include division by a decimal number.

Answers

- 1 **a** 9 **b** 4 **c** 7 **d** 22 **e** 32 **f** 68 **g** 162
 h 720 **i** 27 **j** 1.9 **k** 16.7 **l** 24.7 **m** 68
 n 14.9 **o** 2.95 **p** 5.48 **q** 855 **r** 226
- 2 **a** 459 **b** 523 **c** 1 238 **d** 4 580 **e** 2 876
 f 1 256 **g** 620 **h** 895 **i** 1 426
- 3 **a** 3.9 **b** 2.6 **c** 3.15 **d** 4.61 **e** 3.25
 f 12.5 **g** 9.65 **h** 2.4 **i** 0.73

3.6 Dividing by a decimal B

This activity sheet includes questions involving money and measurements and a selection of word problems. For this activity sheet, students need to write their working and answers in their workbooks.

Answers

- 1 **a** \$24.30 **b** \$32 **c** \$36.60 **d** 31 L
 e 0.6 kg **f** 34 m **g** 37.7 km **h** 3.5 L **i** \$25
- 2 **a** \$14.33 **b** \$17.33 **c** \$4.06 **d** \$19.79
 e \$9.69 **f** \$34.42 **g** \$33.28 **h** \$32.32
 i \$27.41
- 3 16
- 4 8
- 5 \$1.98
- 6 \$8.46

3.7 Mixed calculations with money

This activity sheet begins with 14 warm-up questions involving money which cover the four basic operations. In question 2, students will identify words which refer to one of the four operations: +, −, × or ÷. Allow students time to attempt this without assistance, then discuss their answers and check that they have listed the words in the correct columns.

The second page provides a variety of word problems involving money. Students are asked to underline the key words that tell them the operation to use before starting their calculations. Students should complete these calculations in their workbooks.

Answers

- 1 **a** \$4.75 **b** \$6.04 **c** \$28.60 **d** \$82.34
 e \$2.20 **f** \$11.55 **g** \$3.80 **h** \$35.65
 i \$51.60 **j** \$104.65 **k** \$143.34 **l** \$4.16
 m \$7.32 **n** \$14.20
- 2 +: add, increase, more than, plus, sum, total
 −: decrease, difference, less than, minus, subtract, take away
 ×: lots of, multiply, times
 ÷: divide, share
- 3 **a** \$12.19 **b** \$5.21 **c** \$58.80 **d** \$22.23
 e \$7.45 **f** \$21.30 **g** \$116.10 **h** \$8.65
 i \$22.15 **j** \$47.40 **k** \$12.45 **l** \$24.85
 m \$26.83 **n** \$2.65 **o** \$33.40 **p** \$16.35
 q \$30.12 **r** \$12.15 **s** \$17.19 **t** \$37.04
 u \$8.94 **v** \$54.60 **w** \$329.80 **x** \$87.37
 y \$64.22 **z** \$5.62

EXTENSION IDEAS

Have students work in pairs to make up their own 'maths stories' involving the multiplication or division of decimals. Encourage them to use sensible, real-life contexts – they could use the word problems on the activity sheets in this topic for ideas. When they have finished, they could swap stories with another pair of students and solve them.

Use a calculator to explore the effect of multiplying or dividing decimal numbers by 10, 100 or 1 000. Encourage students to identify and use a quick mental method for these processes.

3.1 Multiplying decimals A

Kirsten bought 0.7 kg of grapes at the local supermarket.
The grapes were priced at \$3.26 per kilogram.
How much did the grapes cost?

To obtain the answer, **multiply 3.26 by 0.7**.



Multiplying two decimals

Step 1: Multiply the two numbers, ignoring the decimal points.

Step 2: Count the number of decimal places in each number in the question.
(How many digits come after the decimal points?)

Step 3: Position the decimal point in your answer so that the
number of decimal places in your answer equals the
total number of decimal places in the question.

EXAMPLES

$$0.8 \times 0.3$$

Step 1: $8 \times 3 = 24$

Step 2: 0.8 has 1 decimal place.
0.3 has 1 decimal place.
Total of 2 decimal places

Step 3: The answer is **0.24**

$$3.26 \times 0.7$$

Step 1:

$$\begin{array}{r} 326 \times \\ 7 \\ \hline 2282 \end{array}$$

Step 2: 3.26 has 2 decimal places.
0.7 has 1 decimal place.
Total of 3 decimal places

Step 3: The answer is **2.282**

1 Multiply these decimals.

a 0.7×0.2

b 0.9×0.6

c 0.4×0.8

d 0.2×0.05

e 0.07×0.4

f 0.05×0.09

3.1 Multiplying decimals A

CONTINUED

g 2.3×0.2

h 3.6×0.7

i 5.9×0.4

j 3.5×0.9

k 5.3×0.5

l 10.3×0.02

m 5.4×0.08

n 0.24×0.3

o 2.93×0.4

p 13.5×0.2

q 2.45×0.8

r 16.3×0.05

s 12.76×0.07

t 18.67×0.4

u 27.6×0.07

Answer the following questions in your workbook.

2 Complete these multiplications.

a Multiply \$13.75 by 5.

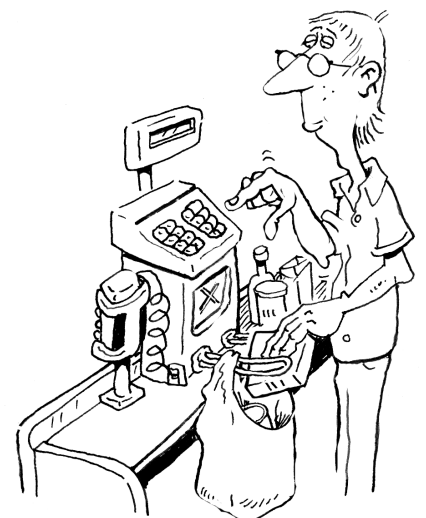
b What is \$9.35 times 4?

c What is 8 lots of \$3.64?

d David has a part-time job.

He earns \$9.40 per hour.

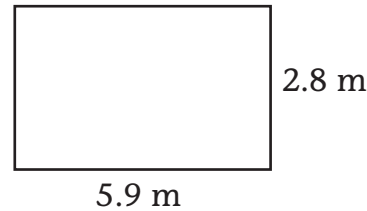
How much will he earn when he works 9 hours?



3.2 Multiplying decimals B

A courtyard is the shape of a rectangle.

Erin multiplied the length by the breadth to calculate the area: $5.9 \text{ m} \times 2.8 \text{ m}$.



Multiplying decimal numbers

Step 1: Multiply the two numbers, ignoring the decimal points.

Step 2: Count the number of decimal places in each number in the question.

5.9 has 1 digit after the decimal point.

2.8 has 1 digit after the decimal point.

There are 2 decimal places altogether.

$$\begin{array}{r} 59 \times \\ 28 \\ \hline 472 \\ 1180 \\ \hline 1652 \end{array}$$

Step 3: Position the decimal point in your answer: 16.52 m^2 .

Answer the following questions in your workbook.

1 Multiply these decimals.

a 2.76×1.5

b 1.45×3.6

c 3.57×2.8

d 8.17×0.35

e 83.4×6.3

f 8.125×7.3

g 7.15×0.89

h 15.75×8.6

i 0.237×5.2

j 94.6×0.51

k 62.5×4.2

l 51.4×8.1

2 Multiply.

a $8.6 \text{ m} \times 1.7$

b $7.4 \text{ kg} \times 1.8$

c $15.9 \text{ L} \times 2.3$

d $\$6.72 \times 1.5$

e $56.3 \text{ kg} \times 2.5$

f $\$15.25 \times 1.2$

g $32.6 \text{ m} \times 2.8$

h $23.4 \text{ L} \times 3.1$

i $\$37.98 \times 5.5$

3 What is the cost of 1.5 kg of apples at $\$2.98$ per kilogram?

4 Calculate the area of a rectangle measuring 2.6 metres by 1.8 metres.

5 Filipa bought 2.5 metres of material costing $\$6.80$ per metre. How much did she pay?

6 Find the cost of 3.8 metres of carpet selling at $\$70.35$ per metre.



3.3 Dividing decimals by whole numbers A

It cost \$2.52 to bake 6 biscuits.
What was the cost of each biscuit?

To obtain the answer, **divide 2.52 by 6**.

When you are dividing a decimal by a whole number, careful setting out is essential. You need to keep the decimal points under one another.



EXAMPLES

$$2.4 \div 4$$

$$\begin{array}{r} 0.6 \\ 4 \overline{)2.4} \end{array}$$

$$2.4 \div 4 = 0.6$$

$$2.52 \div 6$$

$$\begin{array}{r} 0.42 \\ 6 \overline{)2.52} \end{array}$$

$$2.52 \div 6 = 0.42$$

$$0.016 \div 4$$

$$\begin{array}{r} 0.004 \\ 4 \overline{)0.016} \end{array}$$

$$0.016 \div 4 = 0.004$$

1 Complete these divisions.

a $6.8 \div 2$

b $9.6 \div 3$

c $15.5 \div 5$

d $18.6 \div 3$

e $16.8 \div 4$

f $33.6 \div 8$

g $26.4 \div 4$

h $9.42 \div 6$

i $6.48 \div 4$

j $31.55 \div 5$

k $23.04 \div 9$

l $21.84 \div 7$

3.3 Dividing decimals by whole numbers A

CONTINUED

m $0.942 \div 3$

n $0.625 \div 5$

o $0.882 \div 7$

p $0.056 \div 8$

q $0.318 \div 3$

r $0.268 \div 4$

s $0.819 \div 9$

t $2.35 \div 2$

u $18.56 \div 8$

v $4.14 \div 4$

w $17.13 \div 5$

x $8.78 \div 4$

Answer the following questions in your workbook.

2 Complete these divisions.

a $23.1 \div 11$

b $19.5 \div 15$

c $45.6 \div 12$

d $33.8 \div 13$

e $60.2 \div 14$

f $13.68 \div 12$

g $17.76 \div 12$

h $18.45 \div 15$

i $1.26 \div 14$

j $1.44 \div 18$

k $10.4 \div 16$

l $11.57 \div 13$

3 Complete these divisions. Round the answers correct to 2 decimal places.

a $3.7 \div 3$

b $3.92 \div 6$

c $4.12 \div 7$

d $9.05 \div 8$

e $41.3 \div 3$

f $55.6 \div 9$

g $21.5 \div 8$

h $51.2 \div 7$

i $27.93 \div 8$

j $22.55 \div 6$

k $36.56 \div 9$

l $15.63 \div 11$

3.4 Dividing decimals by whole numbers B

\$65.50 was divided equally among 5 girls. How much did each girl receive?

To obtain the answer, we need to divide \$65.50 by 5.

EXAMPLE

$$\$65.50 \div 5$$

Remember, careful setting out is essential.

Keep the decimal points under one another.

The answer is \$13.10.

$$\begin{array}{r} 13.10 \\ 5 \overline{)65.50} \end{array}$$

Answer the following questions in your workbook.

1 Complete the following divisions.

a $\$6.80 \div 4$

b $\$36.50 \div 5$

c $\$127.20 \div 3$

d $4.96 \text{ kg} \div 8$

e $16.8 \text{ litres} \div 3$

f $\$15.68 \div 4$

g $8.6 \text{ metres} \div 4$

h $14.72 \text{ kg} \div 8$

i $324.9 \text{ m} \div 9$

2 Complete the following divisions involving money.

Round your answers to the **nearest cent** (two decimal places).

a $\$8.95 \div 3$

b $\$10.75 \div 6$

c $\$7.85 \div 4$

d $\$20.50 \div 7$

e $\$26.69 \div 3$

f $\$35.95 \div 4$

g $\$42.50 \div 9$

h $\$55.70 \div 8$

i $\$75.85 \div 9$

3 Stacey cut 5.1 metres of rope into 6 pieces of equal length.

How long is each piece?

4 A box containing 9 doughnuts costs \$15.75.

What is the cost of each doughnut?

5 If 8 kilograms of potatoes cost \$14.80,

what is the cost of each kilogram?

6 Travis is paid according to the number of toys

he makes. Yesterday he was paid \$103.95.

He made 9 toys. How much is he paid for each toy?



3.5 Dividing by a decimal A

A drink bottle contains 0.6 litres of water.

How many of these bottles can be filled using 8.4 litres of water?

To obtain the answer, **divide 8.4 by 0.6**. That is, $8.4 \div 0.6$.



Dividing a decimal by another decimal

Step 1: Convert the divisor (the second decimal) to a whole number by moving the decimal point to the right.

Step 2: Move the decimal point in the **dividend** (the first decimal) the **same** number of places in the **same** direction.

Step 3: Perform the division with the new numbers.

EXAMPLE

$$8.4 \div 0.6$$

Step 1: Convert the divisor, **0.6**, to a whole number, **6**, by moving the decimal point 1 place to the right.

Step 2: Move the decimal point in **8.4** one place to the right, giving **84**.

Step 3: Perform the division $84 \div 6$.

The answer is **14**.

1 Complete these divisions.

a $1.8 \div 0.2$

b $1.6 \div 0.4$

c $3.5 \div 0.5$

d $15.4 \div 0.7$

e $19.2 \div 0.6$

f $20.4 \div 0.3$

g $6.48 \div 0.04$

h $14.4 \div 0.02$

i $1.35 \div 0.05$

3.5 Dividing by a decimal A

CONTINUED

j $0.38 \div 0.2$

k $13.36 \div 0.8$

l $12.35 \div 0.5$

m $1.36 \div 0.02$

n $1.341 \div 0.09$

o $0.236 \div 0.08$

p $2.74 \div 0.5$

q $51.3 \div 0.06$

r $67.8 \div 0.3$

Answer the following questions in your workbook.

2 Complete the following divisions.

a $367.2 \div 0.8$

b $261.5 \div 0.5$

c $990.4 \div 0.8$

d $137.4 \div 0.03$

e $143.8 \div 0.05$

f $75.36 \div 0.06$

g $3.1 \div 0.005$

h $3.58 \div 0.004$

i $9.982 \div 0.007$

3 Complete these divisions.

a $4.68 \div 1.2$

b $4.94 \div 1.9$

c $4.725 \div 1.5$

d $8.298 \div 1.8$

e $7.15 \div 2.2$

f $18.75 \div 1.5$

g $15.44 \div 1.6$

h $5.76 \div 2.4$

i $1.825 \div 2.5$



3.6 Dividing by a decimal B

EXAMPLE

Divide \$7.35 by 0.5.

- Convert the **divisor**, 0.5, to a whole number, 5.
- Move the decimal point in the **dividend**, 7.35, the same number of places to give 73.5.
- Calculate the division $73.5 \div 5$.

$$\begin{array}{r} 14.7 \\ 5 \overline{)73.5} \end{array}$$

The answer is **\$14.70**.

Answer the following questions in your workbook.

1 Complete the following divisions.

- | | | |
|---|--|---|
| a $\$12.15 \div 0.5$ | b $\$9.60 \div 0.3$ | c $\$7.32 \div 0.2$ |
| d $15.5 \text{ litres} \div 0.5$ | e $0.42 \text{ kg} \div 0.7$ | f $27.2 \text{ metres} \div 0.8$ |
| g $15.08 \text{ km} \div 0.4$ | h $2.1 \text{ litres} \div 0.6$ | i $\$22.50 \div 0.9$ |

2 Complete the following divisions involving money.

Round your answers to the **nearest cent** (two decimal places).

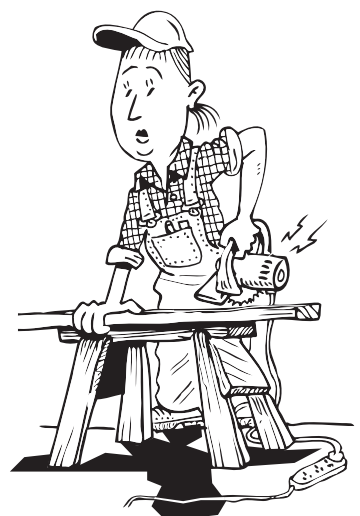
- | | | |
|-----------------------------|-----------------------------|-----------------------------|
| a $\$8.60 \div 0.6$ | b $\$5.20 \div 0.3$ | c $\$3.25 \div 0.8$ |
| d $\$13.85 \div 0.7$ | e $\$8.72 \div 0.9$ | f $\$20.65 \div 0.6$ |
| g $\$29.95 \div 0.9$ | h $\$35.55 \div 1.1$ | i $\$32.89 \div 1.2$ |

3 How many pieces of rope 0.8 metres long can be cut from a length of 12.8 metres?

4 Annaka is cutting shelves from a piece of wood which is 4.8 metres long. Each shelf needs to be 0.6 metres long. How many shelves can she cut?

5 1.5 kg of oranges cost \$2.97. What was the cost of 1 kg of oranges?

6 Last night Brock was paid \$21.15 for working 2.5 hours. How much was he paid for each hour?



3.7 Mixed calculations with money

1 Complete these calculations.

a $\begin{array}{r} \$2.95 \\ + \$1.80 \\ \hline \end{array}$

b $\begin{array}{r} \$4.35 \\ + \$1.69 \\ \hline \end{array}$

c $\begin{array}{r} \$15.75 \\ + \$12.85 \\ \hline \end{array}$

d $\begin{array}{r} \$58.35 \\ + \$23.99 \\ \hline \end{array}$

e $\begin{array}{r} \$5.95 \\ - \$3.75 \\ \hline \end{array}$

f $\begin{array}{r} \$18.80 \\ - \$7.25 \\ \hline \end{array}$

g $\begin{array}{r} \$12.65 \\ - \$8.85 \\ \hline \end{array}$

h $\begin{array}{r} \$73.50 \\ - \$37.85 \\ \hline \end{array}$

i $\$6.45 \times 8$

j $\$14.95 \times 7$

k $\$23.89 \times 6$

l $\$12.48 \div 3$

m $\$36.60 \div 5$

n $\$113.60 \div 8$

Language of operations

There are many ways of expressing the same operation or the same question. For example, 'add \$9 and \$5' and '\$5 more than \$9' both mean $\$9 + \5 .

2 The words in this list each refer to one of the four operations: $+$ $-$ $\times$ $\div$. Write each word in the correct column of the table.

★ add	★ increase	★ more than	★ share	★ times
★ decrease	★ less than	★ multiply	★ subtract	★ sum
★ difference	★ lots of	★ plus	★ take away	★ total
★ divide	★ minus			

+	-	×	÷

3.7 Mixed calculations with money

CONTINUED

- 3** • Underline the words in the following questions which tell you the operation to use: +, −, × **or** ÷.
Refer to the table on the previous page if you need help.
- Then complete these calculations in your workbook.
- | | |
|---|---|
| a Add \$7.63 and \$4.56. | p Divide \$65.40 by 4. |
| b Subtract \$5.29 from \$10.50. | q Decrease \$37.65 by \$7.53. |
| c Multiply \$7.35 by 8. | r Share \$85.05 equally among 7 people. |
| d \$12.98 plus \$9.25 | s Multiply \$11.46 by 1.5. |
| e \$15.30 minus \$7.85 | t Increase \$28.49 by \$8.55. |
| f \$63.90 divided by 3 | u The difference between \$76.89 and \$67.95 |
| g \$19.35 times 6 | v Divide \$436.80 by 8. |
| h The difference between \$24.60 and \$15.95 | w The sum of \$134.60, \$89.95 and \$105.25 |
| i \$6.25 more than \$15.90 | x Decrease \$116.50 by \$29.13. |
| j 12 lots of \$3.95 | y Add \$9.72, \$10.83, \$15.68 and \$27.99. |
| k \$7.50 less than \$19.95 | z By how much is \$32.60 greater than \$26.98? |
| l Increase \$18.35 by \$6.50. | |
| m The sum of \$16.98 and \$9.85 | |
| n Take \$9.35 away from \$12. | |
| o The total of \$9.45, \$12.39 and \$11.56 | |



Teacher's notes

OBJECTIVES

- Understand the concept of a fraction of a whole number or quantity.
- Recognise the numerator and denominator in a fraction.
- Recognise equivalent fractions.
- Reduce a given fraction to its simplest equivalent form.
- Express improper fractions as mixed numerals and vice versa.
- Convert fractions to decimals.

PRIOR KNOWLEDGE

Students should be able to perform the basic operations with whole numbers and understand place value in decimal numbers.

BACKGROUND

Fraction concepts are used in many areas of mathematics, such as chance and measurement, and other subjects such as geography and music.

These activity sheets focus on the basic concepts of fractions, mixed numbers and equivalent fractions. They have been developed for students to complete without using a calculator.

Encourage students to use the terms **numerator** and **denominator** wherever possible.

STARTER ACTIVITY

Tell the students that fractions are used all the time in real life. Challenge them to suggest an area of life in which they are used, and to give an example. If they need hints, you could suggest sporting games, money, time, length measurements and music.

Discuss the relative size of simple fractions. Ask: 'Why is $\frac{1}{4}$ larger than $\frac{1}{6}$?'

For example, when a cake or pizza is divided equally among 6 people, the slices are smaller than if it is shared among 4 people.

ACTIVITY SHEETS

4.1 What is a fraction?

This activity sheet introduces the concept of a fraction of a whole number or quantity, and the terms *numerator* and *denominator*. Students are asked to represent fractions on a diagram and to identify the numerator or denominator of a given fraction.

Answers

- 1 a 3 b 2 c $\frac{2}{3}$
 2 a 8 b 5 c $\frac{5}{8}$
 3 a 10 b 7 c $\frac{7}{10}$
 4 a 4 b 3 c $\frac{3}{4}$
 5 a 8 b 5 c $\frac{5}{8}$
 6 a 12 b 7 c $\frac{7}{12}$
 7 a 6 b 3 c $\frac{3}{6}$
 8 a 5 b 2 c $\frac{2}{5}$
 9 a 4 b 1 c $\frac{1}{4}$
 10 a 8 b 3 c $\frac{3}{8}$
 19 a 2 b 4 c 6 d 5 e 10
 20 a 1 b 2 c 5 d 3 e 5

4.2 Equivalent fractions

This activity sheet develops strategies for finding equivalent fractions, such as multiplying or dividing the numerator and denominator by the same number. Students are then asked to reduce a given fraction to its simplest equivalent form.

Answers

- 2 $\frac{1}{4} = \frac{2}{8} = \frac{3}{12} = \frac{4}{16}$
 3 a $\frac{12}{15}$ b $\frac{12}{16}$ c $\frac{21}{35}$ d $\frac{15}{40}$ e $\frac{8}{28}$ f $\frac{40}{48}$
 4 a 3 b 4 c 5 d 3 e 4 f 5
 5 a 6 b 5 c 12 d 20 e 12 f 20 g 28
 h 24 i 20 j 36 k 35 l 72
 6 a $\frac{2}{3}$ b $\frac{2}{5}$ c $\frac{1}{2}$ d $\frac{3}{5}$ e $\frac{3}{4}$ f $\frac{3}{5}$
 7 a 1 b 3 c 8 d 3 e 5 f 3 g 1 h 10
 8 a $\frac{3}{4}$ b $\frac{1}{3}$ c $\frac{4}{5}$ d $\frac{3}{4}$ e $\frac{3}{4}$ f $\frac{2}{3}$ g $\frac{3}{4}$
 h $\frac{3}{4}$ i $\frac{1}{4}$ j $\frac{7}{10}$ k $\frac{4}{9}$ l $\frac{6}{25}$

Teacher's notes

CONTINUED

4.3 Mixed numbers and improper fractions

A

This activity sheet uses diagrams to introduce equivalent improper fractions and mixed numbers.

Answers

1 a $1\frac{3}{4}, \frac{7}{4}$ b $2\frac{1}{4}, \frac{9}{4}$

2 a $1\frac{7}{8}, \frac{15}{8}$ b $2\frac{3}{8}, \frac{19}{8}$

4.4 Mixed numbers and improper fractions

B

This activity sheet develops mental strategies for changing improper fractions to mixed numbers and vice versa.

Answers

1 a $1\frac{1}{3}$ b $1\frac{1}{4}$ c $2\frac{1}{2}$ d $2\frac{1}{3}$ e $1\frac{1}{5}$ f $3\frac{1}{2}$

g $3\frac{1}{3}$ h $1\frac{3}{4}$ i $1\frac{3}{5}$ j $4\frac{1}{2}$ k $2\frac{2}{5}$ l $2\frac{1}{4}$

m $1\frac{5}{6}$ n $1\frac{1}{8}$ o $3\frac{3}{4}$ p $1\frac{3}{8}$ q $3\frac{1}{4}$ r $1\frac{7}{8}$

s $1\frac{5}{7}$ t $6\frac{1}{4}$ u $2\frac{5}{9}$

2 a $\frac{5}{4}$ b $\frac{5}{2}$ c $\frac{4}{3}$ d $\frac{7}{4}$ e $\frac{7}{3}$ f $\frac{8}{5}$ g $\frac{8}{3}$

h $\frac{13}{5}$ i $\frac{11}{8}$ j $\frac{7}{2}$ k $\frac{10}{3}$ l $\frac{9}{2}$ m $\frac{15}{4}$ n $\frac{23}{10}$

o $\frac{21}{8}$ p $\frac{17}{3}$ q $\frac{23}{5}$ r $\frac{37}{10}$ s $\frac{23}{6}$ t $\frac{35}{8}$ u $\frac{31}{9}$

4.5 Changing fractions to decimals

This activity sheet develops strategies for converting fractions and mixed numbers to decimals.

Answers

1 a 0.1 b 0.6 c 0.53 d 0.95 e 0.573
f 0.197 g 0.103 h 0.9 i 0.87 j 0.212
k 0.09 l 0.089

2 a 0.75 b 0.2 c 0.8 d 0.6 e 0.125
f 0.375 g 0.625 h 0.875 i 0.05 j 0.45
k 0.85 l 0.74 m 0.38 n 0.225 o 0.825

3 a 3.5 b 1.25 c 2.75 d 1.4 e 2.6 f 3.2
g 2.1 h 4.7 i 3.9 j 1.375 k 3.625
l 2.875 m 1.23 n 2.71 o 4.03 p 5.213
q 4.307 r 3.007 s 5.09 t 6.039 u 4.603

4.6 Fractions and repeating decimals

This activity sheet looks at fractions which can be converted to repeating decimals. Some students may need to use a calculator to complete this activity sheet. However, it is not always obvious on a calculator display that the pattern repeats – teacher guidance may be required.

Answers

1 a $0.\dot{6}$ b $0.1\dot{6}$ c $0.\dot{1}$ d $0.\dot{4}$ e $0.\dot{7}$ f $0.\dot{3}\dot{6}$

g $0.0\dot{9}$ h $0.7\dot{2}$ i $0.71428\dot{5}$ j $0.41\dot{6}$

k $0.58\dot{3}$ l $0.\dot{4}2857\dot{1}$ m $0.91\dot{6}$ n $0.08\dot{3}$

o $0.\dot{5}7142\dot{8}$

EXTENSION IDEAS

Provide or work with students to make sets of cards that include matching representations of the same fraction. For example:

$\frac{1}{2}$, 0.5,  , 

You could extend the card sets to include mixed numbers and recurring decimals.

Students could use the card sets for games such as Memory or Go Fish, or they could work in pairs in a timed matching challenge.

4.1 What is a fraction?

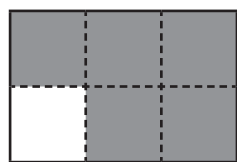
Fractions allow us to show how **whole numbers** are broken into **parts**. For example, games of water polo, football, basketball and many other sports are played in **halves** or **quarters**.

A fraction indicates **part of a whole** number or quantity.



The diagram shows how a rectangular sheet of paper has been folded to form three equal parts.

The shaded part is one-third ($\frac{1}{3}$) of the whole sheet of paper.



The sheet of paper is folded again to form six equal parts. Five parts have been shaded.

Five-sixths ($\frac{5}{6}$) of the rectangle is shaded.

Every fraction has a numerator and a denominator.

- The **numerator** is the top number.
It indicates the number of shaded parts.
 - The **denominator** is the bottom number.
It indicates the total number of equal parts.
- $\frac{5}{6}$

numerator

denominator

Answer questions 1 to 10 in your workbook.

For each diagram, write down:

- the number of equal parts (the denominator)
- the number of shaded parts (the numerator)
- the fraction shaded.

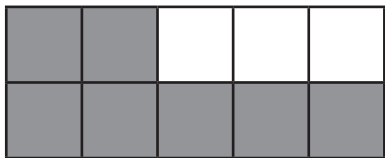
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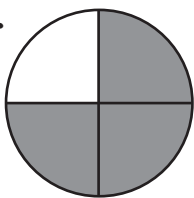
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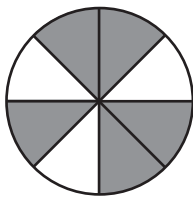
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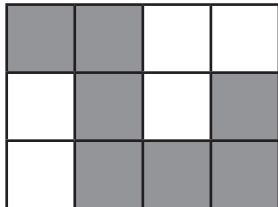
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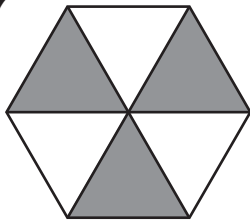
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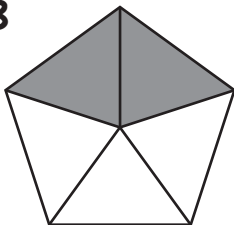
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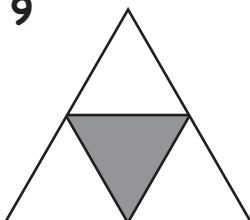
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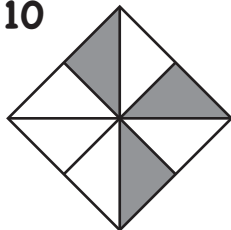
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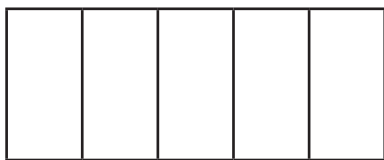
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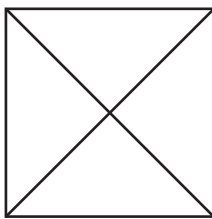
4.1 What is a fraction?

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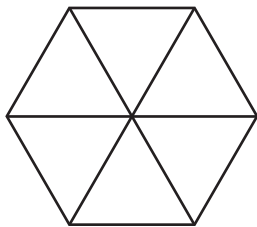
11 Shade $\frac{4}{5}$ of this rectangle.



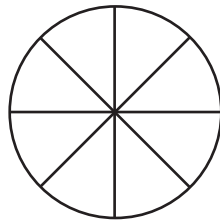
12 Shade $\frac{3}{4}$ of this square.



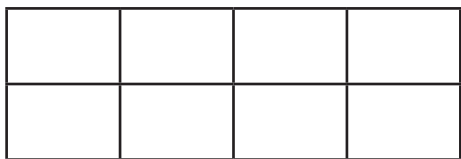
13 Shade $\frac{2}{6}$ of this hexagon.



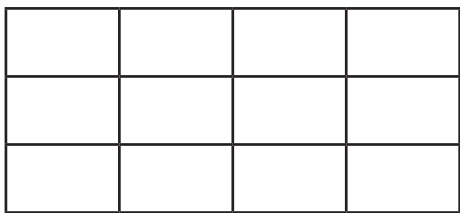
14 Shade $\frac{3}{8}$ of this circle.



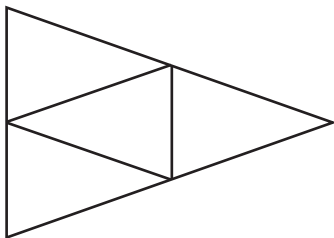
15 Shade $\frac{7}{8}$ of this rectangle.



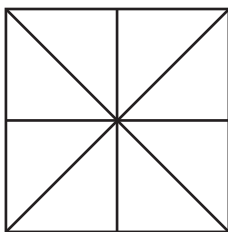
16 Shade $\frac{5}{12}$ of this rectangle.



17 Shade $\frac{3}{4}$ of this triangle.



18 Shade $\frac{5}{8}$ of this square.



19 What is the value of the **denominator** in these fractions?

- a $\frac{1}{2}$
2
- b $\frac{3}{4}$

- c $\frac{5}{6}$

- d $\frac{2}{5}$

- e $\frac{7}{10}$

20 What is the value of the **numerator** in these fractions?

- a $\frac{1}{6}$

- b $\frac{2}{3}$

- c $\frac{5}{8}$

- d $\frac{3}{5}$

- e $\frac{5}{12}$

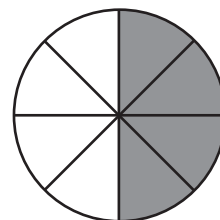
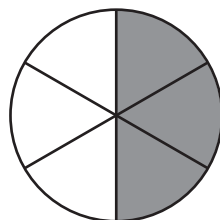
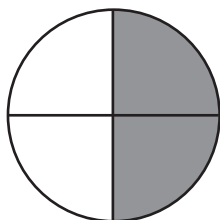
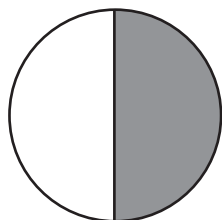
4.2 Equivalent fractions

Kamal got an answer of **2 quarters** to a fraction problem.

Neelam said the answer was **3 sixths**. Do they have the same value?

Here are four family-size pizzas. They have been cut in different ways.

Half of each pizza has been shaded.

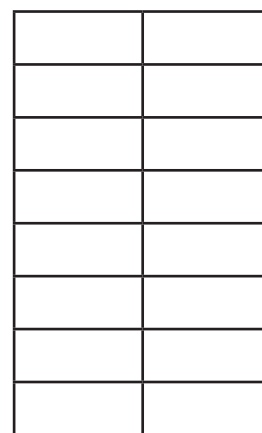
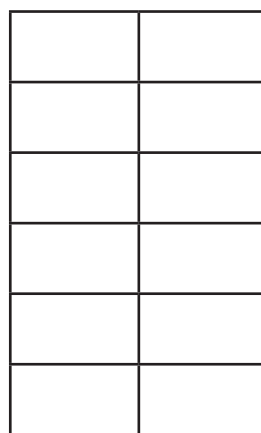
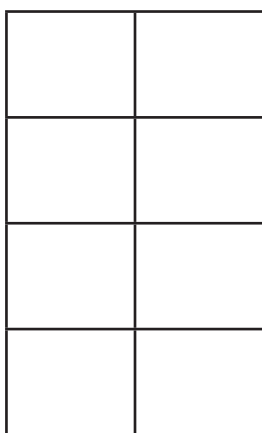
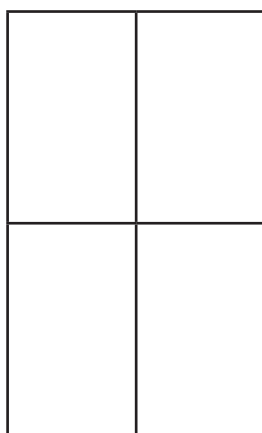


The diagrams above show a set of **equivalent fractions**: $\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8}$

Equivalent fractions have the **same value**.

So, 2 quarters and 3 sixths have the same value.

- 1** The following four rectangles are the same size. They have been divided in different ways. On each diagram, shade one quarter ($\frac{1}{4}$) of the rectangle.



- 2** Complete this set of equivalent fractions: $\frac{1}{4} = \frac{\quad}{8} = \frac{\quad}{12} = \frac{\quad}{16}$

We make equivalent fractions by multiplying **both** the **numerator** (top number) and the **denominator** (bottom number) by the **same number**.

EXAMPLES

$$\frac{3}{5} = \frac{3 \times 2}{5 \times 2} = \frac{6}{10}$$

$$\frac{2}{3} = \frac{2 \times 5}{3 \times 5} = \frac{10}{15}$$

- 3** Complete the following to form equivalent fractions.

a $\frac{4}{5} = \frac{4 \times 3}{5 \times 3} = \frac{\quad}{\quad}$

b $\frac{3}{4} = \frac{3 \times 4}{4 \times 4} = \frac{\quad}{\quad}$

c $\frac{3}{5} = \frac{3 \times 7}{5 \times 7} = \frac{\quad}{\quad}$

d $\frac{3}{8} = \frac{3 \times 5}{8 \times 5} = \frac{\quad}{\quad}$

e $\frac{2}{7} = \frac{2 \times 4}{7 \times 4} = \frac{\quad}{\quad}$

f $\frac{5}{6} = \frac{5 \times 8}{6 \times 8} = \frac{\quad}{\quad}$

4.2 Equivalent fractions

CONTINUED

4 Find the number to replace the $\triangle$ in each of the following.

a $\frac{1}{2} = \frac{1 \times \triangle}{2 \times \triangle} = \frac{3}{6}$

$\triangle = \underline{\quad}$

b $\frac{2}{3} = \frac{2 \times \triangle}{3 \times \triangle} = \frac{8}{12}$

$\triangle = \underline{\quad}$

c $\frac{3}{4} = \frac{3 \times \triangle}{4 \times \triangle} = \frac{15}{20}$

$\triangle = \underline{\quad}$

d $\frac{5}{6} = \frac{5 \times \triangle}{6 \times \triangle} = \frac{15}{18}$

$\triangle = \underline{\quad}$

e $\frac{4}{7} = \frac{4 \times \triangle}{7 \times \triangle} = \frac{16}{28}$

$\triangle = \underline{\quad}$

f $\frac{7}{8} = \frac{7 \times \triangle}{8 \times \triangle} = \frac{35}{40}$

$\triangle = \underline{\quad}$

5 Complete the following to form equivalent fractions.

a $\frac{1}{2} = \frac{\quad}{12}$

b $\frac{1}{4} = \frac{\quad}{20}$

c $\frac{2}{3} = \frac{\quad}{18}$

d $\frac{5}{6} = \frac{\quad}{24}$

e $\frac{1}{3} = \frac{4}{\quad}$

f $\frac{3}{4} = \frac{15}{\quad}$

g $\frac{3}{7} = \frac{12}{\quad}$

h $\frac{7}{8} = \frac{21}{\quad}$

i $\frac{4}{9} = \frac{\quad}{45}$

j $\frac{5}{6} = \frac{30}{\quad}$

k $\frac{5}{9} = \frac{\quad}{63}$

l $\frac{3}{8} = \frac{27}{\quad}$

We can **simplify** or reduce a fraction if we can **divide** the numerator and the denominator by the **same number**.

EXAMPLES

$$\frac{5}{10} = \frac{5 \div 5}{10 \div 5} = \frac{1}{2}$$

$$\frac{12}{15} = \frac{12 \div 3}{15 \div 3} = \frac{4}{5}$$

6 Complete these divisions to simplify each fraction.

a $\frac{6}{9} = \frac{6 \div 3}{9 \div 3} = \underline{\quad}$

b $\frac{4}{10} = \frac{4 \div 2}{10 \div 2} = \underline{\quad}$

c $\frac{6}{12} = \frac{6 \div 6}{12 \div 6} = \underline{\quad}$

d $\frac{9}{15} = \frac{9 \div 3}{15 \div 3} = \underline{\quad}$

e $\frac{15}{20} = \frac{15 \div 5}{20 \div 5} = \underline{\quad}$

f $\frac{12}{20} = \frac{12 \div 4}{20 \div 4} = \underline{\quad}$

7 Complete the following equivalent fractions.

a $\frac{3}{6} = \frac{\quad}{2}$

b $\frac{6}{10} = \frac{\quad}{5}$

c $\frac{14}{16} = \frac{7}{\quad}$

d $\frac{12}{18} = \frac{2}{\quad}$

e $\frac{16}{20} = \frac{4}{\quad}$

f $\frac{9}{12} = \frac{\quad}{4}$

g $\frac{10}{30} = \frac{\quad}{3}$

h $\frac{70}{100} = \frac{7}{\quad}$

8 Write these fractions in their simplest form.

a $\frac{6}{8} = \underline{\quad}$

b $\frac{3}{9} = \underline{\quad}$

c $\frac{8}{10} = \underline{\quad}$

d $\frac{9}{12} = \underline{\quad}$

e $\frac{15}{20} = \underline{\quad}$

f $\frac{12}{18} = \underline{\quad}$

g $\frac{18}{24} = \underline{\quad}$

h $\frac{33}{44} = \underline{\quad}$

i $\frac{8}{32} = \underline{\quad}$

j $\frac{70}{100} = \underline{\quad}$

k $\frac{28}{63} = \underline{\quad}$

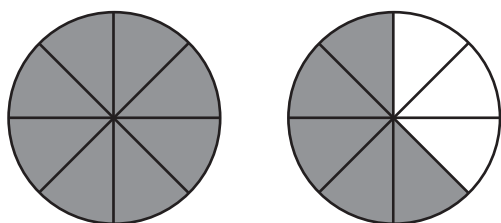
l $\frac{24}{100} = \underline{\quad}$

4.3 Mixed numbers and improper fractions A

Julie and Rhys bought two pizzas. Each pizza was cut into 8 pieces.

They ate 13 pieces or thirteen eighths ($\frac{13}{8}$).

The pieces eaten have been shaded in this diagram.

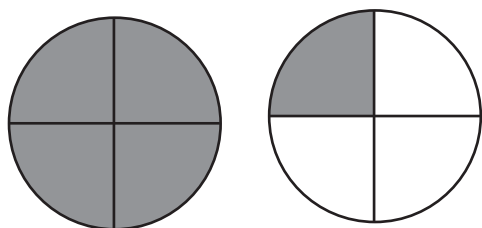


They ate one whole pizza and five eighths of the other.

$1 + \frac{5}{8}$ is written as $1\frac{5}{8}$.

$1\frac{5}{8}$ is called a **mixed number**. It contains a whole number and a fraction.

$\frac{13}{8}$ is called an **improper fraction**. The numerator is larger than the denominator.



This diagram shows two pizzas.

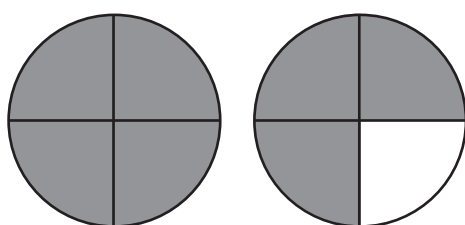
These two pizzas have been cut into **quarters**.

The pieces that have been eaten are shaded.

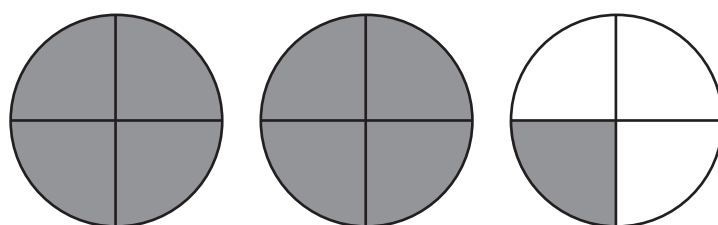
The fraction shaded can be written as a mixed number ($1\frac{1}{4}$) or as an improper fraction ($\frac{5}{4}$).

- 1** Write the fractions shaded in the following pizza diagrams.
Write your answers both as mixed numbers and improper fractions.

a _____

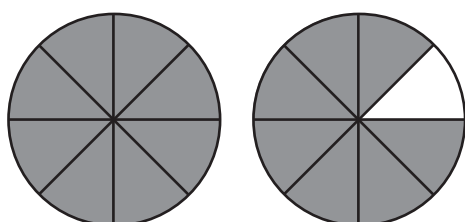


b _____

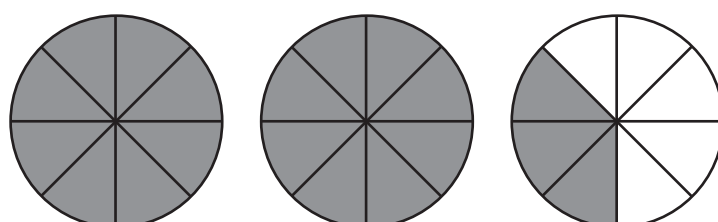


- 2** The following diagrams show pizzas cut into **eighths**.
Write the fractions shaded both as mixed numbers and improper fractions.

a _____



b _____



4.4 Mixed numbers and improper fractions B

An improper fraction can be written as a mixed number.

For example, five thirds of this diagram is shaded.



$$\frac{5}{3} = 1\frac{2}{3}$$

EXAMPLE

Write the improper fraction $\frac{5}{3}$ as a mixed number.

To change an improper fraction to a mixed number

- Ask yourself: *How many times does 3 divide into 5?*
- There is **1 lot of 3** in 5 and a **remainder of 2**.

So $\frac{5}{3} = 1\frac{2}{3}$



- 1** Change the following improper fractions to mixed numbers.
You could draw diagrams in your workbook to help.

a $\frac{4}{3} =$ _____

b $\frac{5}{4} =$ _____

c $\frac{5}{2} =$ _____

d $\frac{7}{3} =$ _____

e $\frac{6}{5} =$ _____

f $\frac{7}{2} =$ _____

g $\frac{10}{3} =$ _____

h $\frac{7}{4} =$ _____

i $\frac{8}{5} =$ _____

j $\frac{9}{2} =$ _____

k $\frac{12}{5} =$ _____

l $\frac{9}{4} =$ _____

m $\frac{11}{6} =$ _____

n $\frac{9}{8} =$ _____

o $\frac{15}{4} =$ _____

p $\frac{11}{8} =$ _____

q $\frac{13}{4} =$ _____

r $\frac{15}{8} =$ _____

s $\frac{12}{7} =$ _____

t $\frac{25}{4} =$ _____

u $\frac{23}{9} =$ _____

4.4 Mixed numbers and improper fractions B

CONTINUED

EXAMPLE

Change $2\frac{3}{5}$ to an improper fraction.

To change a mixed number to an improper fraction



$$\begin{aligned} 2\frac{3}{5} &= \frac{2 \times 5}{5} + \frac{3}{5} \\ &= \frac{10}{5} + \frac{3}{5} \\ &= \frac{13}{5} \end{aligned}$$

The whole number 2 is equal to $2 \times 5 = 10$ fifths.

Add the extra 3 fifths
to give $\frac{13}{5}$ (thirteen fifths).

2 Change these mixed numbers to improper fractions.

a $1\frac{1}{4} = \underline{\hspace{2cm}}$

b $2\frac{1}{2} = \underline{\hspace{2cm}}$

c $1\frac{1}{3} = \underline{\hspace{2cm}}$

d $1\frac{3}{4} = \underline{\hspace{2cm}}$

e $2\frac{1}{3} = \underline{\hspace{2cm}}$

f $1\frac{3}{5} = \underline{\hspace{2cm}}$

g $2\frac{2}{3} = \underline{\hspace{2cm}}$

h $2\frac{3}{5} = \underline{\hspace{2cm}}$

i $1\frac{3}{8} = \underline{\hspace{2cm}}$

j $3\frac{1}{2} = \underline{\hspace{2cm}}$

k $3\frac{1}{3} = \underline{\hspace{2cm}}$

l $4\frac{1}{2} = \underline{\hspace{2cm}}$

m $3\frac{3}{4} = \underline{\hspace{2cm}}$

n $2\frac{3}{10} = \underline{\hspace{2cm}}$

o $2\frac{5}{8} = \underline{\hspace{2cm}}$

p $5\frac{2}{3} = \underline{\hspace{2cm}}$

q $4\frac{3}{5} = \underline{\hspace{2cm}}$

r $3\frac{7}{10} = \underline{\hspace{2cm}}$

s $3\frac{5}{6} = \underline{\hspace{2cm}}$

t $4\frac{3}{8} = \underline{\hspace{2cm}}$

u $3\frac{4}{9} = \underline{\hspace{2cm}}$



4.5 Changing fractions to decimals

People use fractions in conversations every day. We use decimals with money and measurement. The ability to recognise common fractions in their decimal form is a useful skill to have.

EXAMPLES

Fractions written as 10ths, 100ths or 1000ths

Tenths have **one** decimal place:

$$\frac{3}{10} = 0.3$$

Hundredths have **two** decimal places:

$$\frac{47}{100} = 0.47$$

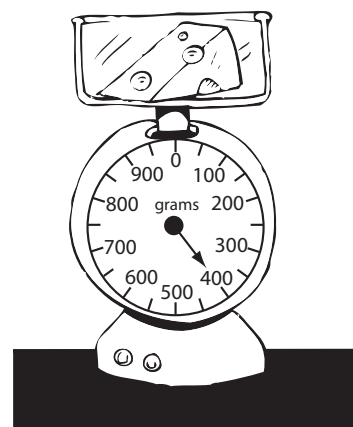
$$\frac{7}{100} = 0.07$$

Thousandths have **three** decimal places:

$$\frac{153}{1000} = 0.153$$

$$\frac{37}{1000} = 0.037$$

$$\frac{9}{1000} = 0.009$$



1 Write as decimals.

a $\frac{1}{10} =$ _____

b $\frac{6}{10} =$ _____

c $\frac{53}{100} =$ _____

d $\frac{95}{100} =$ _____

e $\frac{573}{1000} =$ _____

f $\frac{197}{1000} =$ _____

g $\frac{103}{1000} =$ _____

h $\frac{9}{10} =$ _____

i $\frac{87}{100} =$ _____

j $\frac{212}{1000} =$ _____

k $\frac{9}{100} =$ _____

l $\frac{89}{1000} =$ _____

EXAMPLES

Fractions can be written as divisions.

$$\frac{1}{4} = 1 \div 4 \\ = 0.25$$

$$\frac{0.25}{4 \overline{)1.00}}$$

$$\frac{2}{5} = 2 \div 5 \\ = 0.4$$

$$\frac{0.4}{5 \overline{)2.0}}$$

2 Use division to change these fractions to decimals.

a $\frac{3}{4} =$ _____

b $\frac{1}{5} =$ _____

c $\frac{4}{5} =$ _____

4.5 Changing fractions to decimals

CONTINUED

d $\frac{3}{5} =$ _____

e $\frac{1}{8} =$ _____

f $\frac{3}{8} =$ _____

g $\frac{5}{8} =$ _____

h $\frac{7}{8} =$ _____

i $\frac{1}{20} =$ _____

j $\frac{9}{20} =$ _____

k $\frac{17}{20} =$ _____

l $\frac{37}{50} =$ _____

m $\frac{19}{50} =$ _____

n $\frac{9}{40} =$ _____

o $\frac{33}{40} =$ _____

EXAMPLES

Writing mixed numbers as decimals

$$\begin{aligned} 1\frac{1}{2} &= 1 + \frac{1}{2} \\ &= 1 + 0.5 \\ &= 1.5 \end{aligned}$$

Notice that the **whole number** is written **in front of** the decimal point.

$$\begin{aligned} 2\frac{3}{10} &= 2 + \frac{3}{10} \\ &= 2 + 0.3 \\ &= 2.3 \end{aligned}$$

3 Write the following mixed numbers as decimals.

a $3\frac{1}{2} =$ _____

b $1\frac{1}{4} =$ _____

c $2\frac{3}{4} =$ _____

d $1\frac{2}{5} =$ _____

e $2\frac{3}{5} =$ _____

f $3\frac{1}{5} =$ _____

g $2\frac{1}{10} =$ _____

h $4\frac{7}{10} =$ _____

i $3\frac{9}{10} =$ _____

j $1\frac{3}{8} =$ _____

k $3\frac{5}{8} =$ _____

l $2\frac{7}{8} =$ _____

m $1\frac{23}{100} =$ _____

n $2\frac{71}{100} =$ _____

o $4\frac{3}{100} =$ _____

p $5\frac{213}{1000} =$ _____

q $4\frac{307}{1000} =$ _____

r $3\frac{7}{1000} =$ _____

s $5\frac{9}{100} =$ _____

t $6\frac{39}{1000} =$ _____

u $4\frac{603}{1000} =$ _____



4.6 Fractions and repeating decimals

Kristina wanted to convert $\frac{1}{3}$ to a decimal.

She wrote the fraction as $1 \div 3$ and attempted the division.

$$\begin{array}{r} 0.333\ 33\ \dots \\ 3 \overline{) 1.000\ 00\ \dots} \end{array}$$

This division will never stop.

The answer $0.333\ 33\ \dots$ is a **repeating** or recurring decimal.

It is written as $0.\dot{3}$.

The dot above the digit 3 indicates that it is repeating.



Travis wanted to convert $\frac{5}{6}$ to a decimal by division.

$$\frac{5}{6} = 5 \div 6$$

$$\begin{array}{r} 0.833\ 3\ \dots \\ 6 \overline{) 5.000\ 0\ \dots} \end{array}$$

Again, the decimal repeats.

It is written as $0.8\dot{3}$ (only the 3 is repeating).



To show that a set or pattern of digits is repeating, place a dot above the first and last repeating digits: $0.273\ 273\ 273\ \dots = 0.\dot{2}7\dot{3}$.

1 Use division to change these fractions to decimals.

You will need to do your working on a sheet of paper or in your workbook.

a $\frac{2}{3} =$ _____

b $\frac{1}{6} =$ _____

c $\frac{1}{9} =$ _____

d $\frac{4}{9} =$ _____

e $\frac{7}{9} =$ _____

f $\frac{4}{11} =$ _____

g $\frac{1}{11} =$ _____

h $\frac{8}{11} =$ _____

i $\frac{5}{7} =$ _____

j $\frac{5}{12} =$ _____

k $\frac{7}{12} =$ _____

l $\frac{3}{7} =$ _____

m $\frac{11}{12} =$ _____

n $\frac{1}{12} =$ _____

o $\frac{4}{7} =$ _____

Teacher's notes

OBJECTIVES

- Add fractions and mixed numbers.
- Subtract fractions and mixed numbers.
- Multiply fractions and mixed numbers.
- Divide using fractions and mixed numbers.
- Calculate a fraction of a given quantity.

PRIOR KNOWLEDGE

Students should understand the concept of a fraction and be familiar with the terms numerator and denominator. They should be able to perform the basic operations with whole numbers, without using a calculator. They should be able to reduce a fraction to its lowest equivalent form and express mixed numbers as improper fractions and vice versa.

BACKGROUND

The following activity sheets have been developed for students to complete without using a calculator. They cover the four basic operations with simple fractions and mixed numbers. The activity sheets extending over two pages can readily be split and each page used separately if required.

STARTER ACTIVITY

Conduct a class quiz to consolidate the students' skills at converting a given fraction to different forms, such as improper fractions to mixed numbers and then to decimal fractions. You could also ask them to reduce fractions to simplest form, and give equivalencies such as '2 fifths is how many tenths?'

ACTIVITY SHEETS

5.1 Adding fractions A

This activity sheet develops students' skills in adding fractions with the same denominator. These skills are then extended to mixed numbers.

Answers

$$\begin{array}{llllll}
 1 \text{ a } \frac{5}{7} & \text{b } \frac{3}{5} & \text{c } \frac{3}{7} & \text{d } \frac{4}{8} = \frac{1}{2} & \text{e } \frac{7}{9} & \text{f } \frac{6}{8} = \frac{3}{4} \\
 \text{g } \frac{6}{11} & \text{h } \frac{4}{10} = \frac{2}{5} & \text{i } \frac{5}{5} = 1 & \text{j } \frac{5}{10} = \frac{1}{2} & \text{k } \frac{8}{8} = 1 \\
 \text{l } \frac{10}{10} = 1 & \text{m } \frac{6}{5} = 1\frac{1}{5} & \text{n } \frac{11}{9} = 1\frac{2}{9} & \text{o } \frac{15}{10} = 1\frac{1}{2} \\
 2 \text{ a } 2\frac{3}{5} & \text{b } 3\frac{2}{3} & \text{c } 3\frac{4}{5} & \text{d } 3\frac{2}{4} = 3\frac{1}{2} & \text{e } 2\frac{4}{8} = 2\frac{1}{2}
 \end{array}$$

$$\begin{array}{llllll}
 \text{f } 3\frac{5}{9} & \text{g } 4\frac{5}{7} & \text{h } 5\frac{4}{10} = 5\frac{2}{5} & \text{i } 5\frac{6}{8} = 5\frac{3}{4} & \text{j } 4 \\
 \text{k } 3 & \text{l } 5 & \text{m } 4\frac{6}{5} = 5\frac{1}{5} & \text{n } 4\frac{12}{8} = 5\frac{1}{2} \\
 \text{o } 5\frac{9}{7} = 6\frac{2}{7}
 \end{array}$$

5.2 Adding fractions B

This activity sheet first deals with the addition of simple fractions that have different denominators. It then looks at the addition of mixed numbers. The activity sheet concludes with some word problems requiring addition. For this activity sheet, students need to write their working and answers in their workbooks.

Answers

$$\begin{array}{llllll}
 1 \text{ a } \frac{5}{6} & \text{b } \frac{13}{20} & \text{c } \frac{19}{20} & \text{d } \frac{29}{30} & \text{e } \frac{23}{24} & \text{f } \frac{29}{30} & \text{g } \frac{29}{35} \\
 \text{h } \frac{17}{18} & \text{i } \frac{17}{12} = 1\frac{5}{12} & \text{j } \frac{17}{14} = 1\frac{3}{14} & \text{k } \frac{17}{15} = 1\frac{2}{15} \\
 \text{l } \frac{27}{20} = 1\frac{7}{20} & \text{m } \frac{37}{24} = 1\frac{13}{24} & \text{n } \frac{43}{30} = 1\frac{13}{30} \\
 \text{o } \frac{31}{24} = 1\frac{7}{24} & \text{p } \frac{79}{70} = 1\frac{9}{70} \\
 2 \text{ a } 3\frac{4}{5} & \text{b } 3\frac{11}{15} & \text{c } 5\frac{17}{20} & \text{d } 3\frac{11}{18} & \text{e } 5\frac{9}{14} & \text{f } 4\frac{3}{14} \\
 \text{g } 4\frac{13}{24} & \text{h } 5\frac{7}{20} \\
 3 \frac{11}{12} & 4 \ 1\frac{5}{12} \text{ hours} & 5 \frac{31}{40} & 6 \ 6\frac{1}{4}
 \end{array}$$

5.3 Subtracting fractions A

This activity sheet develops students' skills in subtracting fractions with the same denominator. These skills are then extended to mixed numbers.

Answers

$$\begin{array}{llllll}
 1 \text{ a } \frac{2}{4} = \frac{1}{2} & \text{b } \frac{1}{3} & \text{c } \frac{1}{5} & \text{d } \frac{3}{7} & \text{e } \frac{4}{6} = \frac{2}{3} & \text{f } \frac{2}{9} \\
 \text{g } \frac{2}{8} = \frac{1}{4} & \text{h } \frac{6}{9} = \frac{2}{3} & \text{i } \frac{2}{5} & \text{j } \frac{3}{9} = \frac{1}{3} & \text{k } \frac{6}{12} = \frac{1}{2} \\
 \text{l } \frac{6}{14} = \frac{3}{7} & \text{m } \frac{4}{10} = \frac{2}{5} & \text{n } \frac{10}{15} = \frac{2}{3} & \text{o } \frac{10}{20} = \frac{1}{2} \\
 2 \text{ a } 1\frac{2}{5} & \text{b } 1\frac{2}{8} = 1\frac{1}{4} & \text{c } 2\frac{1}{3} & \text{d } 1\frac{2}{4} = 1\frac{1}{2} \\
 \text{e } 1\frac{3}{9} = 1\frac{1}{3} & \text{f } 2\frac{2}{7} & \text{g } \frac{6}{8} = \frac{3}{4} & \text{h } 1\frac{1}{4} & \text{i } \frac{2}{3} \\
 \text{j } 1\frac{3}{5} & \text{k } 1\frac{4}{7} & \text{l } 1\frac{1}{2} & \text{m } 1\frac{1}{2} & \text{n } 1\frac{1}{3} & \text{o } 1\frac{3}{5}
 \end{array}$$

Teacher's notes

CONTINUED

5.4 Subtracting fractions B

This activity sheet first deals with the subtraction of simple fractions that have different denominators. It then looks at subtracting mixed numbers. The activity sheet concludes with some word problems which require subtraction. For this activity sheet, students need to write their working and answers in their workbooks.

Answers

$$\begin{array}{llllll}
 1 \text{ a } \frac{1}{6} & \text{b } \frac{1}{10} & \text{c } \frac{7}{20} & \text{d } \frac{1}{12} & \text{e } \frac{7}{24} & \text{f } \frac{5}{24} \\
 \text{g } \frac{9}{18} = \frac{1}{2} & \text{h } \frac{4}{32} = \frac{1}{8} & \text{i } \frac{8}{20} = \frac{2}{5} & \text{j } \frac{1}{30} & & \\
 \text{k } \frac{6}{40} = \frac{3}{20} & \text{l } \frac{2}{48} = \frac{1}{24} & & & & \\
 2 \text{ a } 1\frac{5}{12} & \text{b } 1\frac{1}{20} & \text{c } 1\frac{11}{20} & \text{d } 1\frac{5}{21} & \text{e } 2\frac{1}{2} & \text{f } 1\frac{3}{10} \\
 \text{g } 2\frac{5}{8} & \text{h } 1\frac{7}{12} & \text{i } \frac{8}{15} & \text{j } 1\frac{7}{10} & \text{k } 1\frac{7}{8} & \text{l } 2\frac{33}{35} \\
 3 \frac{1}{10} & 4 \frac{1}{6} & 5 \frac{3}{4} \text{ litres} & 6 \frac{1}{10} \text{ kg} & &
 \end{array}$$

5.5 Multiplying fractions

This activity sheet first develops students' skills in multiplying simple fractions. The second page extends their skills to multiplying mixed numbers and then provides some word problems which require multiplication. For this activity sheet, students need to write their working and answers in their workbooks. You may wish to show students how to cancel common factors before they multiply. In the author's experience, weaker students are less likely to make an error using the pen and paper method shown on the activity sheets (that is, multiply, **then** simplify the answer).

Answers

$$\begin{array}{llllll}
 1 \text{ a } \frac{1}{12} & \text{b } \frac{3}{8} & \text{c } \frac{2}{15} & \text{d } \frac{2}{25} & \text{e } \frac{4}{15} & \text{f } \frac{4}{10} = \frac{2}{5} \\
 \text{g } \frac{9}{30} = \frac{3}{10} & \text{h } \frac{12}{20} = \frac{3}{5} & \text{i } \frac{6}{40} = \frac{3}{20} & \text{j } \frac{14}{24} = \frac{7}{12} & & \\
 \text{k } \frac{20}{40} = \frac{1}{2} & \text{l } \frac{18}{30} = \frac{3}{5} & \text{m } \frac{15}{90} = \frac{1}{6} & \text{n } \frac{40}{108} = \frac{10}{27} & & \\
 \text{o } \frac{45}{80} = \frac{9}{16} & \text{p } \frac{30}{84} = \frac{5}{14} & & & & \\
 2 \text{ a } \frac{7}{12} & \text{b } 1\frac{1}{5} & \text{c } \frac{9}{16} & \text{d } 1 & \text{e } 1\frac{2}{3} & \text{f } 3\frac{1}{2} & \text{g } 4\frac{1}{5} \\
 \text{h } 3\frac{1}{3} & \text{i } 3\frac{1}{3} & \text{j } 2 & \text{k } 6 & \text{l } 7\frac{7}{10} & \text{m } 10\frac{5}{8} & \\
 \text{n } 12 & \text{o } 16 & \text{p } 7\frac{3}{5} & & & & \\
 3 \frac{1}{2} \text{ kg} & 4 \frac{3}{8} & 5 \frac{9}{20} & 6 \frac{3}{8} & 7 \frac{12}{8} \text{ km} & &
 \end{array}$$

5.6 Division with fractions

This activity sheet first develops students' skills in dividing by a simple fraction. The second page extends their skills to division with mixed numbers and then provides some word problems which require division. For this activity sheet, students need to write their working and answers in their workbooks.

Answers

$$\begin{array}{llllll}
 1 \text{ a } \frac{2}{3} & \text{b } \frac{1}{2} & \text{c } \frac{5}{2} = 2\frac{1}{2} & \text{d } \frac{8}{9} & \text{e } \frac{15}{8} = 1\frac{7}{8} & \\
 \text{f } \frac{4}{5} & \text{g } \frac{5}{8} & \text{h } \frac{3}{4} & \text{i } \frac{5}{3} = 1\frac{2}{3} & \text{j } \frac{21}{24} = \frac{7}{8} & \\
 \text{k } \frac{32}{15} = 2\frac{2}{15} & \text{l } 1\frac{1}{6} & \text{m } 2\frac{2}{3} & \text{n } 1\frac{4}{21} & \text{o } \frac{5}{12} & \\
 \text{p } 1\frac{3}{7} & & & & & \\
 2 \text{ a } 9 & \text{b } 3\frac{1}{2} & \text{c } 1\frac{7}{8} & \text{d } 3\frac{1}{2} & \text{e } \frac{8}{15} & \text{f } \frac{9}{32} & \text{g } \frac{9}{14} \\
 \text{h } 1\frac{11}{16} & \text{i } 1\frac{5}{9} & \text{j } 2 & \text{k } 3 & \text{l } \frac{3}{4} & \text{m } \frac{16}{25} & \text{n } 1\frac{8}{9} \\
 \text{o } 1\frac{9}{10} & \text{p } 3 & & & & & \\
 3 \frac{7}{8} & 4 \text{ } 32 & 5 \frac{3}{4} \text{ hours} & 6 \text{ } 5 & & &
 \end{array}$$

5.7 Fractions of quantities

This activity sheet develops strategies for calculating fractions of quantities in the contexts of money and measurement. The examples at the top of the first page suggest one method that the author has found most students are comfortable with applying.

Answers

$$\begin{array}{llllll}
 1 \text{ a } \$10 & \text{b } \$3 & \text{c } \$5 & \text{d } \$6 & \text{e } \$9 & \text{f } \$5 \\
 \text{g } \$45 & \text{h } \$32 & \text{i } \$23 & \text{j } \$8 & \text{k } \$16 & \text{l } \$500 \\
 \text{m } \$108 & \text{n } \$45 & \text{o } \$26 & \text{p } \$570 & \text{q } \$840 & \\
 \text{r } \$100 & \text{s } \$4.20 & \text{t } \$5.70 & \text{u } \$4.10 & \text{v } \$3 & \\
 \text{w } \$8.40 & \text{x } \$15.60 & & & & \\
 2 \text{ a } 82 \text{ km} & \text{b } 107 \text{ kg} & \text{c } 70 \text{ L} & \text{d } 16 \text{ m} & & \\
 \text{e } 30 \text{ tonnes} & \text{f } 450 \text{ L} & \text{g } 57 \text{ km} & \text{h } 66 \text{ mL} & & \\
 \text{i } 68 \text{ kg} & \text{j } 99 \text{ L} & \text{k } 240 \text{ tonnes} & \text{l } 238 \text{ km} & &
 \end{array}$$

EXTENSION IDEAS

Students could work in pairs to compile a fractions quiz that features calculations with all four operations. You could encourage them to use a real-life context such as a timber yard/hardware store or supermarket. Have one student use a calculator to check the answers, and the other student use pen and paper methods.

5.1 Adding fractions A

Adrian scored $\frac{3}{5}$ of his team's goals last season.

Carlos scored $\frac{1}{5}$ of the team's goals.

What fraction of the team's goals did these two players score?

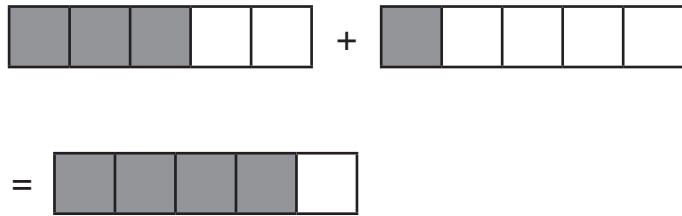
We need to add the fractions $\frac{3}{5}$ and $\frac{1}{5}$.



EXAMPLE

When fractions have the **same** (or common) denominator, the numerators may be added. The denominator **does not** change.

$$\begin{aligned}\frac{3}{5} + \frac{1}{5} \\&= \frac{3+1}{5} \\&= \frac{4}{5}\end{aligned}$$



1 Complete these additions. Simplify your answers where possible.

a $\frac{4}{7} + \frac{1}{7}$

= _____

b $\frac{2}{5} + \frac{1}{5}$

= _____

c $\frac{2}{7} + \frac{1}{7}$

= _____

d $\frac{3}{8} + \frac{1}{8}$

= _____

e $\frac{5}{9} + \frac{2}{9}$

= _____

f $\frac{5}{8} + \frac{1}{8}$

= _____

g $\frac{5}{11} + \frac{1}{11}$

= _____

h $\frac{3}{10} + \frac{1}{10}$

= _____

i $\frac{3}{5} + \frac{2}{5}$

= _____

j $\frac{4}{10} + \frac{1}{10}$

= _____

k $\frac{3}{8} + \frac{5}{8}$

= _____

l $\frac{7}{10} + \frac{3}{10}$

= _____

m $\frac{4}{5} + \frac{2}{5}$

= _____

n $\frac{7}{9} + \frac{4}{9}$

= _____

o $\frac{9}{10} + \frac{6}{10}$

= _____

5.1 Adding fractions A

CONTINUED

EXAMPLE

$$2\frac{3}{5} + 1\frac{1}{5}$$

When you add mixed numbers, add the whole numbers first. Then add the fractions.

$$\begin{aligned} 2\frac{3}{5} + 1\frac{1}{5} &= 2 + 1 + \frac{3}{5} + \frac{1}{5} \\ &= 3 + \frac{4}{5} \\ &= 3\frac{4}{5} \end{aligned}$$

2 Add these mixed numbers. Simplify your answers where possible.

a $1\frac{2}{5} + 1\frac{1}{5}$

= _____

b $2\frac{1}{3} + 1\frac{1}{3}$

= _____

c $2\frac{3}{5} + 1\frac{1}{5}$

= _____

d $2\frac{1}{4} + 1\frac{1}{4}$

= _____

e $1\frac{1}{8} + 1\frac{3}{8}$

= _____

f $2\frac{3}{9} + 1\frac{2}{9}$

= _____

g $3\frac{2}{7} + 1\frac{3}{7}$

= _____

h $3\frac{1}{10} + 2\frac{3}{10}$

= _____

i $3\frac{1}{8} + 2\frac{5}{8}$

= _____

j $2\frac{1}{4} + 1\frac{3}{4}$

= _____

k $1\frac{2}{3} + 1\frac{1}{3}$

= _____

l $3\frac{3}{5} + 1\frac{2}{5}$

= _____

m $3\frac{2}{5} + 1\frac{4}{5}$

= _____

n $2\frac{7}{8} + 2\frac{5}{8}$

= _____

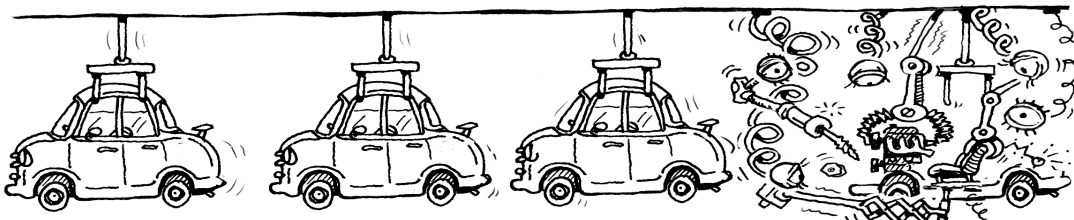
o $3\frac{4}{7} + 2\frac{5}{7}$

= _____

= _____

= _____

= _____



5.2 Adding fractions B

EXAMPLE

$$\frac{2}{5} + \frac{1}{3}$$

Adding fractions with different denominators

Step 1: Multiply the denominators.

Step 2: Cross multiply, starting at the top left.
Write these products on the top line.

Step 3: Add the products on the top line.

Step 4: Simplify your answer where possible.

$$\frac{2}{5} + \frac{1}{3}$$

$$\begin{array}{r} 5 \times 3 \\ 2 \times 3 + 1 \times 5 \\ \hline 15 \end{array}$$

$$= \frac{6 + 5}{15}$$

$$= \frac{11}{15}$$

Answer the following questions in your workbook.

1 Complete these additions.

a $\frac{1}{2} + \frac{1}{3}$

b $\frac{1}{4} + \frac{2}{5}$

c $\frac{3}{4} + \frac{1}{5}$

d $\frac{2}{3} + \frac{3}{10}$

e $\frac{5}{8} + \frac{1}{3}$

f $\frac{4}{5} + \frac{1}{6}$

g $\frac{2}{5} + \frac{3}{7}$

h $\frac{4}{9} + \frac{1}{2}$

i $\frac{3}{4} + \frac{2}{3}$

j $\frac{5}{7} + \frac{1}{2}$

k $\frac{1}{3} + \frac{4}{5}$

l $\frac{3}{4} + \frac{3}{5}$

m $\frac{7}{8} + \frac{2}{3}$

n $\frac{5}{6} + \frac{3}{5}$

o $\frac{5}{8} + \frac{2}{3}$

p $\frac{3}{7} + \frac{7}{10}$

2 Add these mixed numbers.

Remember, add the whole numbers first. Then add the fractions.

a $1\frac{3}{5} + 2\frac{1}{5}$

b $2\frac{1}{3} + 1\frac{2}{5}$

c $2\frac{3}{5} + 3\frac{1}{4}$

d $2\frac{1}{9} + 1\frac{1}{2}$

e $3\frac{1}{2} + 2\frac{1}{7}$

f $2\frac{5}{7} + 1\frac{1}{2}$

g $2\frac{7}{8} + 1\frac{2}{3}$

h $1\frac{3}{4} + 3\frac{3}{5}$

3 What is the sum of $\frac{2}{3}$ and $\frac{1}{4}$?

4 A science theory test went for $\frac{3}{4}$ hour. A practical test went for $\frac{2}{3}$ hour. What was the total length of the two tests, as a mixed number?

5 Sebastian spent $\frac{3}{8}$ of his pocket money on Saturday. He spent $\frac{2}{5}$ of his money on Sunday. What fraction of his pocket money did Sebastian spend on these two days?

6 Ellyse set out on a bushwalk. She walked $3\frac{1}{2}$ km before she stopped for a rest. She then walked a further $2\frac{3}{4}$ km. How many kilometres did she walk altogether?

5.3 Subtracting fractions A

Caitlin and Markus ate $\frac{7}{8}$ of a pizza.

Markus ate $\frac{3}{8}$ of the pizza.

What fraction of the pizza did Caitlin eat?

When two fractions have the **same** (or common) denominator, the numerators may be subtracted.

The denominator **does not** change.



EXAMPLE

$$\begin{aligned}\frac{7}{8} - \frac{3}{8} &= \frac{7-3}{8} \\ &= \frac{4}{8} \text{ which simplifies to } \frac{1}{2}\end{aligned}$$

1 Complete these subtractions. Simplify your answers where possible.

a $\frac{3}{4} - \frac{1}{4}$

= _____

b $\frac{2}{3} - \frac{1}{3}$

= _____

c $\frac{3}{5} - \frac{2}{5}$

= _____

d $\frac{6}{7} - \frac{3}{7}$

= _____

e $\frac{5}{6} - \frac{1}{6}$

= _____

f $\frac{7}{9} - \frac{5}{9}$

= _____

g $\frac{5}{8} - \frac{3}{8}$

= _____

h $\frac{8}{9} - \frac{2}{9}$

= _____

i $\frac{4}{5} - \frac{2}{5}$

= _____

j $\frac{7}{9} - \frac{4}{9}$

= _____

k $\frac{11}{12} - \frac{5}{12}$

= _____

l $\frac{9}{14} - \frac{3}{14}$

= _____

m $\frac{7}{10} - \frac{3}{10}$

= _____

n $\frac{14}{15} - \frac{4}{15}$

= _____

o $\frac{17}{20} - \frac{7}{20}$

= _____

5.3 Subtracting fractions A

CONTINUED

When you subtract **mixed numbers**, subtract the whole numbers first.
Then subtract the fractions.

EXAMPLES

$$\begin{aligned} 3\frac{7}{8} - 1\frac{3}{8} \\ &= (3 - 1) + \left(\frac{7}{8} - \frac{3}{8}\right) \\ &= 2 + \frac{7-3}{8} \\ &= 2\frac{4}{8} \\ &= 2\frac{1}{2} \end{aligned}$$

$$\begin{aligned} 4\frac{2}{5} - 1\frac{4}{5} \\ &= (4 - 1) + \left(\frac{2}{5} - \frac{4}{5}\right) \\ &= 3 + \left(-\frac{2}{5}\right) \text{ or } 3 - \frac{2}{5} \\ &= 2 + 1 - \frac{2}{5} \text{ or } 2 + \frac{5}{5} - \frac{2}{5} \\ &= 2\frac{3}{5} \end{aligned}$$

2 Subtract these mixed numbers. Simplify your answers where possible.

a $2\frac{3}{5} - 1\frac{1}{5}$

= _____

b $2\frac{5}{8} - 1\frac{3}{8}$

= _____

c $3\frac{2}{3} - 1\frac{1}{3}$

= _____

d $2\frac{3}{4} - 1\frac{1}{4}$

= _____

e $3\frac{7}{9} - 2\frac{4}{9}$

= _____

f $3\frac{4}{7} - 1\frac{2}{7}$

= _____

g $1\frac{7}{8} - 1\frac{1}{8}$

= _____

h $3 - 1\frac{3}{4}$

= _____

i $2\frac{1}{3} - 1\frac{2}{3}$

= _____

j $3\frac{1}{5} - 1\frac{3}{5}$

= _____

k $4\frac{2}{7} - 2\frac{5}{7}$

= _____

l $4\frac{1}{4} - 2\frac{3}{4}$

= _____

m $3\frac{3}{8} - 1\frac{7}{8}$

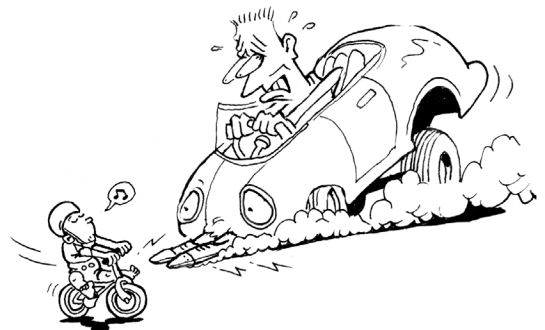
= _____

n $5\frac{1}{9} - 3\frac{7}{9}$

= _____

o $4\frac{3}{10} - 2\frac{7}{10}$

= _____



5.4 Subtracting fractions B

EXAMPLE

$$\frac{4}{5} - \frac{1}{3}$$

$$\frac{4}{5} - \frac{1}{3}$$

To subtract fractions with different denominators

Step 1: Multiply the denominators.

$$\frac{\quad}{5 \times 3}$$

Step 2: Cross multiply, starting at the top left.
Write these products on the top line.

$$\frac{4 \times 3 - 1 \times 5}{15}$$

Step 3: Subtract the products on the top line.

$$= \frac{12 - 5}{15}$$

Step 4: Simplify your answer where possible.

$$= \frac{7}{15}$$

Answer the following questions in your workbook.

1 Complete these subtractions.

a $\frac{1}{2} - \frac{1}{3}$

b $\frac{3}{5} - \frac{1}{2}$

c $\frac{3}{5} - \frac{1}{4}$

d $\frac{3}{4} - \frac{2}{3}$

e $\frac{5}{8} - \frac{1}{3}$

f $\frac{7}{8} - \frac{2}{3}$

g $\frac{2}{3} - \frac{1}{6}$

h $\frac{7}{8} - \frac{3}{4}$

i $\frac{9}{10} - \frac{1}{2}$

j $\frac{7}{10} - \frac{2}{3}$

k $\frac{9}{10} - \frac{3}{4}$

l $\frac{7}{8} - \frac{5}{6}$

2 Subtract these mixed numbers.

Remember, subtract the whole numbers first. Then subtract the fractions.

a $2\frac{3}{4} - 1\frac{1}{3}$

b $2\frac{4}{5} - 1\frac{3}{4}$

c $3\frac{3}{4} - 2\frac{1}{5}$

d $3\frac{4}{7} - 2\frac{1}{3}$

e $3\frac{2}{3} - 1\frac{1}{6}$

f $2\frac{7}{10} - 1\frac{2}{5}$

g $4\frac{7}{8} - 2\frac{1}{4}$

h $3\frac{1}{3} - 1\frac{3}{4}$

i $2\frac{1}{5} - 1\frac{2}{3}$

j $4\frac{1}{2} - 2\frac{4}{5}$

k $3\frac{3}{4} - 1\frac{7}{8}$

l $5\frac{4}{5} - 2\frac{6}{7}$

3 Find the difference between $\frac{3}{5}$ and $\frac{1}{2}$.

4 The petrol tank in Sarah's car was two thirds full. She used one half of a tank of petrol on a journey. What fraction of a tank of petrol remains?

5 A bucket contained $6\frac{1}{2}$ litres of liquid fertiliser. $2\frac{3}{4}$ litres of this fertiliser were used on a garden. How much is left in the bucket?

6 The mass of potatoes in a bag was $7\frac{1}{2}$ kg.
Tyson used $3\frac{2}{5}$ kg during the week. What mass of potatoes was left?

5.5 Multiplying fractions

At the local cycling club, one third of the members are juniors.

Half of the junior members are female.

What fraction of the club's members are female juniors?

To find the answer, we need to **multiply** $\frac{1}{2}$ and $\frac{1}{3}$.



When multiplying fractions, you are finding a fraction of a fraction.

Multiplying two fractions

Step 1: Multiply the numerators together.

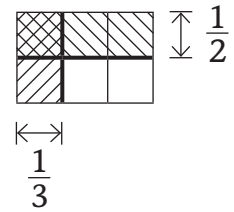
Step 2: Multiply the denominators together.

Step 3: Simplify your answer, if possible.

EXAMPLES

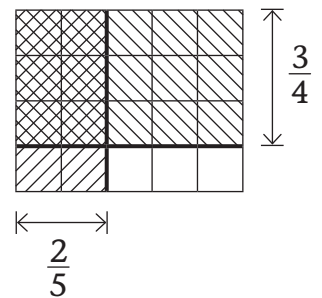
$$\frac{1}{2} \times \frac{1}{3} = \frac{1}{2} \text{ of } \frac{1}{3}$$

$$\begin{aligned} \frac{1}{2} \times \frac{1}{3} &= \frac{1 \times 1}{2 \times 3} \\ &= \frac{1}{6} \end{aligned}$$



$$\frac{3}{4} \times \frac{2}{5} = \frac{3}{4} \text{ of } \frac{2}{5}$$

$$\begin{aligned} \frac{3}{4} \times \frac{2}{5} &= \frac{3 \times 2}{4 \times 5} \\ &= \frac{6}{20} \\ &= \frac{3}{10} \end{aligned}$$



1 Multiply these fractions. Remember to simplify your answers, if possible.

a $\frac{1}{3} \times \frac{1}{4}$

= _____

b $\frac{1}{2} \times \frac{3}{4}$

= _____

c $\frac{1}{3} \times \frac{2}{5}$

= _____

d $\frac{2}{5} \times \frac{1}{5}$

= _____

e $\frac{2}{3} \times \frac{2}{5}$

= _____

f $\frac{1}{2} \times \frac{4}{5}$

= _____

g $\frac{1}{3} \times \frac{9}{10}$

= _____

h $\frac{3}{4} \times \frac{4}{5}$

= _____

i $\frac{2}{5} \times \frac{3}{8}$

= _____

j $\frac{2}{3} \times \frac{7}{8}$

= _____

k $\frac{4}{5} \times \frac{5}{8}$

= _____

l $\frac{9}{10} \times \frac{2}{3}$

= _____

m $\frac{3}{10} \times \frac{5}{9}$

= _____

n $\frac{5}{12} \times \frac{8}{9}$

= _____

o $\frac{5}{8} \times \frac{9}{10}$

= _____

p $\frac{6}{7} \times \frac{5}{12}$

= _____

5.5 Multiplying fractions

CONTINUED

You can use the same method when multiplying mixed numbers.
First change the mixed numbers to improper fractions.

EXAMPLES

$$\begin{aligned}\frac{3}{5} \times 3\frac{2}{3} &= \frac{3}{5} \times \frac{11}{3} \\ &= \frac{3 \times 11}{5 \times 3} \\ &= \frac{33}{15} \text{ or } \frac{11}{5} \\ &= 2\frac{1}{5}\end{aligned}$$

$$\begin{aligned}2\frac{1}{2} \times 1\frac{3}{4} &= \frac{5}{2} \times \frac{7}{4} \\ &= \frac{5 \times 7}{2 \times 4} \\ &= \frac{35}{8} \\ &= 4\frac{3}{8}\end{aligned}$$

Answer the following questions in your workbook.

2 Complete these multiplications.

a $\frac{1}{3} \times 1\frac{3}{4}$

b $\frac{3}{4} \times 1\frac{3}{5}$

c $1\frac{7}{8} \times \frac{3}{10}$

d $1\frac{3}{5} \times \frac{5}{8}$

e $1\frac{1}{4} \times 1\frac{1}{3}$

f $1\frac{1}{2} \times 2\frac{1}{3}$

g $1\frac{1}{5} \times 3\frac{1}{2}$

h $2\frac{1}{2} \times 1\frac{1}{3}$

i $2\frac{2}{3} \times 1\frac{1}{4}$

j $1\frac{3}{5} \times 1\frac{1}{4}$

k $1\frac{4}{5} \times 3\frac{1}{3}$

l $1\frac{3}{4} \times 4\frac{2}{5}$

m $3\frac{2}{5} \times 3\frac{1}{8}$

n $2\frac{1}{4} \times 5\frac{1}{3}$

o $2\frac{2}{5} \times 6\frac{2}{3}$

p $4\frac{3}{4} \times 1\frac{3}{5}$

3 A packet contains $\frac{3}{4}$ kg of flour when it is full. $\frac{2}{3}$ of the contents have been used. How much flour was used? Answer as a fraction of 1 kg.

4 Brett had $\frac{3}{4}$ of a family size pizza.
He gave each of his two brothers half of this.
What fraction of the pizza was each brother given?

5 In a fishing club, $\frac{3}{5}$ of the members are male.
 $\frac{3}{4}$ of the male members are single.
What fraction of the club's members are single males?

6 What is the product of $2\frac{1}{2}$ and $1\frac{3}{4}$?

7 Jemma can walk $5\frac{1}{2}$ km each hour.
How far can she walk in $2\frac{1}{4}$ hours?

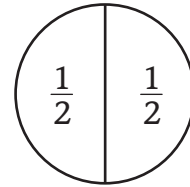
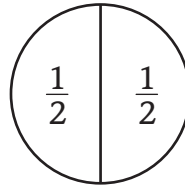
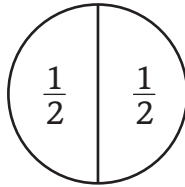


5.6 Division with fractions

Three oranges were each cut in half. How many pieces does this give?

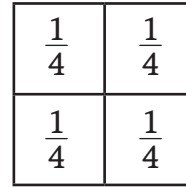
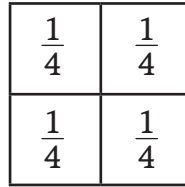
How many halves in 3?

$$3 \div \frac{1}{2} = 3 \times 2 = 6$$



How many quarters in 2?

$$2 \div \frac{1}{4} = 2 \times 4 = 8$$



When dividing by a fraction, **multiply** by its **reciprocal**.

The reciprocal is the fraction 'turned upside down'.

The reciprocal of $\frac{1}{4}$ is $\frac{4}{1} = 4$. The reciprocal of $\frac{2}{5}$ is $\frac{5}{2} = 2\frac{1}{2}$.

EXAMPLES

$$\begin{aligned} \frac{1}{2} \div \frac{1}{4} &= \frac{1}{2} \times \frac{4}{1} \\ &= \frac{1 \times 4}{2 \times 1} \\ &= \frac{4}{2} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \frac{3}{4} \div \frac{2}{5} &= \frac{3}{4} \times \frac{5}{2} \\ &= \frac{3 \times 5}{4 \times 2} \\ &= \frac{15}{8} \\ &= 1\frac{7}{8} \end{aligned}$$

Answer these questions in your workbook.

1 Complete these divisions.

Remember to write each answer in its simplest form.

a $\frac{1}{3} \div \frac{1}{2}$

b $\frac{1}{4} \div \frac{1}{2}$

c $\frac{1}{2} \div \frac{1}{5}$

d $\frac{2}{3} \div \frac{3}{4}$

e $\frac{3}{4} \div \frac{2}{5}$

f $\frac{3}{5} \div \frac{3}{4}$

g $\frac{3}{8} \div \frac{3}{5}$

h $\frac{3}{5} \div \frac{4}{5}$

i $\frac{5}{8} \div \frac{3}{8}$

j $\frac{3}{4} \div \frac{6}{7}$

k $\frac{4}{5} \div \frac{3}{8}$

l $\frac{7}{8} \div \frac{3}{4}$

m $\frac{4}{5} \div \frac{3}{10}$

n $\frac{5}{6} \div \frac{7}{10}$

o $\frac{3}{8} \div \frac{9}{10}$

p $\frac{5}{6} \div \frac{7}{12}$

5.6 Division with fractions

CONTINUED

You can use the same method when dividing with mixed numbers.
First change the mixed numbers to improper fractions.

EXAMPLES

$$\begin{aligned} 2\frac{1}{4} \div \frac{1}{2} &= \frac{9}{4} \div \frac{1}{2} \\ &= \frac{9}{4} \times \frac{2}{1} \\ &= \frac{9 \times 2}{4 \times 1} \\ &= \frac{18}{4} = \frac{9}{2} \\ &= 4\frac{1}{2} \end{aligned}$$

$$\begin{aligned} 1\frac{3}{4} \div 3\frac{1}{2} &= \frac{7}{4} \div \frac{7}{2} \\ &= \frac{7}{4} \times \frac{2}{7} \\ &= \frac{7 \times 2}{4 \times 7} \\ &= \frac{14}{28} \text{ or } \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

Answer these questions in your workbook.

2 Complete these divisions.

a $2\frac{1}{4} \div \frac{1}{4}$

b $2\frac{1}{3} \div \frac{2}{3}$

c $1\frac{1}{2} \div \frac{4}{5}$

d $2\frac{1}{2} \div \frac{5}{7}$

e $\frac{2}{3} \div 1\frac{1}{4}$

f $\frac{3}{8} \div 1\frac{1}{3}$

g $1\frac{1}{2} \div 2\frac{1}{3}$

h $2\frac{1}{4} \div 1\frac{1}{3}$

i $3\frac{1}{2} \div 2\frac{1}{4}$

j $2\frac{1}{2} \div 1\frac{1}{4}$

k $4\frac{1}{2} \div 1\frac{1}{2}$

l $1\frac{1}{3} \div 1\frac{7}{9}$

m $1\frac{3}{5} \div 2\frac{1}{2}$

n $2\frac{5}{6} \div 1\frac{1}{2}$

o $4\frac{3}{4} \div 2\frac{1}{2}$

p $4\frac{4}{5} \div 1\frac{3}{5}$

3 Divide $\frac{3}{4}$ by $\frac{2}{5}$.

4 It takes $\frac{3}{8}$ of a bottle of soft drink to fill a large glass.
How many of these glasses can be filled by
12 bottles of soft drink?

5 Naomi practised her guitar for the same length of time
each day. After 5 days, she had practised for a total
of $8\frac{3}{4}$ hours. For how long did she practise each day?

6 A piece of rope $7\frac{1}{2}$ metres long is cut into $1\frac{1}{2}$ metre lengths.
How many pieces of rope will be produced?



5.7 Fractions of quantities

Markus had \$60. He gave one third of this to Nico and two fifths to Carla.
How much did Nico and Carla receive?

EXAMPLES

Calculate $\frac{1}{3}$ of \$60.

Divide by the denominator

$$60 \div 3 = 20$$

$$\frac{1}{3} \text{ of } \$60 = \$20$$

Nico received \$20.

Calculate $\frac{2}{5}$ of \$60.

Step 1: Divide by the denominator.

$$60 \div 5 = 12 \quad \left(\frac{1}{5} \text{ of } 60 = 12\right)$$

Step 2: Multiply by the numerator.

$$12 \times 2 = 24$$

$$\frac{2}{5} \text{ of } \$60 = \$24$$

Carla received \$24.

1 Calculate:

a $\frac{1}{2}$ of \$20

= _____

= _____

b $\frac{1}{4}$ of \$12

= _____

= _____

c $\frac{1}{3}$ of \$15

= _____

= _____

d $\frac{1}{5}$ of \$30

= _____

= _____

e $\frac{1}{6}$ of \$54

= _____

= _____

f $\frac{1}{8}$ of \$40

= _____

= _____

g $\frac{1}{5}$ of \$225

= _____

= _____

h $\frac{1}{7}$ of \$224

= _____

= _____

i $\frac{1}{6}$ of \$138

= _____

= _____

j $\frac{2}{3}$ of \$12

k $\frac{2}{5}$ of \$40

l $\frac{5}{6}$ of \$600

5.7 Fractions of quantities

CONTINUED

m $\frac{4}{5}$ of \$135

n $\frac{3}{8}$ of \$120

o $\frac{2}{7}$ of \$91

p $\frac{3}{4}$ of \$760

q $\frac{7}{8}$ of \$960

r $\frac{4}{9}$ of \$225

s $\frac{2}{3}$ of \$6.30

t $\frac{3}{4}$ of \$7.60

u $\frac{2}{5}$ of \$10.25

v $\frac{5}{8}$ of \$4.80

w $\frac{4}{9}$ of \$18.90

x $\frac{6}{11}$ of \$28.60

Answer these questions in your workbook.

2 Calculate.

a $\frac{1}{4}$ of 328 km

b $\frac{1}{3}$ of 321 kg

c $\frac{1}{6}$ of 420 L

d $\frac{1}{8}$ of 128 m

e $\frac{2}{3}$ of 45 tonnes

f $\frac{3}{4}$ of 600 L

g $\frac{3}{4}$ of 76 km

h $\frac{2}{5}$ of 165 mL

i $\frac{4}{7}$ of 119 kg

j $\frac{3}{8}$ of 264 L

k $\frac{5}{6}$ of 288 tonnes

l $\frac{7}{8}$ of 272 km

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